

Existence and analytic behaviour of Jost solutions for Schrödinger operators on a one-dimensional lattice with varying spectral multiplicity

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Abstract

We study scattering theory for a class of matrix-valued Schrödinger operators on a one-dimensional lattice. In our model, the free Hamiltonian is given by a variation of the discrete Laplacian, and we consider a matrix potential satisfying a condition of exponential decay. After proving the existence of the Jost solutions, we use them to construct the scattering matrix, whose dimension varies depending on the energy.

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Notation

- Given $N \in \mathbb{N}$ and $k \in \{1, \dots, N\}$, we write $e_k \in \mathbb{C}^N$ for the k -th standard basis vector, i.e.

$$e_k = (\delta_{kj})_{j=1, \dots, N},$$

denoting by δ the Kronecker delta.

- Given two positive integers $N, M \in \mathbb{N}$, we write $\mathcal{M}_{N,M}(\mathbb{C})$ for the set of complex $N \times M$ -matrices.
- Given $N \in \mathbb{N}$, we set $\mathcal{M}_N(\mathbb{C}) := \mathcal{M}_{N,M}(\mathbb{C})$ and write $\mathrm{GL}_N(\mathbb{C})$ for the set of invertible elements in $\mathcal{M}_N(\mathbb{C})$.
- Given $N \in \mathbb{N}$, we write $\mathbf{1}_N$ for the neutral element of the group $\mathrm{GL}_N(\mathbb{C})$.
- We define the complex half-planes $\mathbb{C}_{\pm} := \{z \in \mathbb{C} : \pm \mathrm{Im}(z) > 0\}$.

Introduction

We study scattering theory for a class of matrix-valued Schrödinger operators on $\mathbb{Z}$ of form

$$H = H_0 + V,$$

where the free Hamiltonian H_0 is an energy shift of the discrete Laplace operator composed with the left multiplication by some invertible matrix A and the potential V is a self-adjoint matrix multiplication operator obeying an exponential growth condition (cf. Section 1 for a detailed definition). This model is widely used in the context of low-energy phenomena in solid-state physics.

The case $A = \mathbf{1}_L$ has already been studied in [BFGS22] and [BFNS24], where the corresponding scattering matrix has been constructed. The novelty of this paper is that – contrary to the results in [BFGS22] and [BFNS24] – for matrix A that is not a scalar multiply of a unitary matrix, in our model, the dimension of the scattering matrix varies depending on the energy. Such models have recently been of interest, for example, in [APRR24], although there, in a similar model, just the case of a potential V supported in a single point has been studied.

Structure of the paper:

In Section 1, we will introduce our model and prove that the free Hamiltonian H_0 has purely absolutely continuous spectrum with varying spectral multiplicity (Proposition ??).

In Section ??, we study the generalized eigenfunctions of H_0 and prove the existence of Jost solutions, i.e. generalized eigenfunctions of H with prescribed behavior at $\pm\infty$ (Corollary ??).

In Section ?? we construct the scattering matrix of our model. For this, in Section ??, we prove that, for almost every energy E , the dimension of the space of bounded generalized eigenfunctions of H_0 and H coincide (Theorem ??, Proposition ??). We then, in Section ??, use this fact to construct an almost everywhere defined matrix-valued function

$$\sigma(H_0) \ni E \mapsto S^E$$

(Definition ??). We also prove that, where S^E is defined, it is unitary (Proposition ??) and continuous/analytic (Corollary ??).

1 The model

Given $L \in \mathbb{N}$ and an invertible normal matrix $A \in \text{GL}_L(\mathbb{C})$, we consider the free Hamiltonian H_0 on $\mathcal{H} := \ell^2(\mathbb{Z}, \mathbb{C}^L)$ defined by

$$(H_0\phi)(n) := A^*\phi(n+1) + A\phi(n-1).$$

Considering a map

$$V : \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C})$$

with

$$V(n) = V(n)^* \quad \forall n \in \mathbb{Z},$$

for $\phi \in \mathcal{H}$, we set

$$(V\phi)(n) := V(n)\phi(n)$$

and consider the interaction Hamiltonian

$$H := H_0 + V.$$

The goal of this paper is to calculate the scattering matrix for the pair (H_0, H) .

Definition 1.1. [τ_A and the Strict Growth Condition and] We use the symbol τ_A to denote the next quantity:

$$\tau_A := \eta_A + \sqrt{\eta_A^2 - 1} \quad \text{with} \quad \eta_A := \|A\| \cdot \|A^{-1}\|.$$

We say that V satisfies the strict growth condition if there exists $\varepsilon > 1$, with $\varepsilon \geq \tau_A$, such that

$$\sum_{n \in \mathbb{Z}} \|V(n)\| \varepsilon^{|n|} < \infty. \quad (1)$$

Throughout this paper, we assume that V satisfies the growth condition.

Notice that, for $A \in \text{GL}_L(\mathbb{C})$, one has $\eta_A \geq 1$ and therefore also $\tau_A \geq 1$. Further, $\tau_A = 1$, if and only if $\eta_A = 1$, which holds true if and only if A is unitary (notice that $\|A\| = 1 = \|A^{-1}\|$ implies that A is an isometry: $\|Ax\| \leq \|x\|$ and $\|x\| = \|A^{-1}Ax\| \leq \|Ax\|$, for every x). In the case that $\tau_A > 1$ we can simply take $\varepsilon = \tau_A$.

Remark 1.2. In the case that $\tau_A = 1$, condition (1) becomes

$$\sum_{n \in \mathbb{Z}} \|V(n)\| < \infty,$$

which is precisely the case studied in [BFGS22] and [BFNS24]. Many of our results also hold in this case but, for some of our results, we have to assume that (1) holds.

We start by simplifying our problem through some suitable transformations. Since scattering theory is invariant under unitary transformations, we want to show that after a certain unitary transformation, we can, without loss of generality, assume that the matrix A is positive. For this, we consider the polar decomposition of A , i.e. a unitary matrix $U \in \text{U}(\mathbb{C}^L)$ such that

$$A = U|A| = |A|U,$$

where the second equality holds since A is a normal matrix. We now consider the unitary operator $\tilde{U} \in \mathcal{U}(\mathcal{H})$ defined by

$$(\tilde{U}\phi)(n) = U^n\phi(n), \quad n \in \mathbb{Z}, \phi \in \mathcal{H}.$$

Then, for $n \in \mathbb{Z}$ and $\phi \in \mathcal{H}$, we have

$$\begin{aligned} \left(\tilde{U}^{-1}H_0\tilde{U}\phi\right)(n) &= U^{-n} [A^*U^{n+1}\phi(n+1) + AU^{n-1}\phi(n-1)] \\ &= UA^*\phi(n+1) + U^{-1}A\phi(n-1) \\ &= |A|\phi(n+1) + |A|\phi(n-1) \end{aligned}$$

and

$$\left(\tilde{U}^{-1}V\tilde{U}\phi\right)(n) = U^{-n}V(n)U^n\phi(n) = (U^*)^nV(n)U^n\phi(n).$$

These calculations show that by replacing A with $|A|$ and $V(n)$ with $(U^*)^nV(n)U^n$, $n \in \mathbb{Z}$, we can, without loss of generality, assume that **A is a positive semi-definite Hermitian matrix** $A > 0$.

Notice that

$$\tau_A = \tau_{|A|}$$

and

$$\|(U^*)^nV(n)U^n\| = \|V(n)\| \quad \forall n \in \mathbb{Z},$$

so that condition (1) remains unchanged under the above transformation.

Since **A is a positive semi-definite Hermitian matrix**, it is diagonalizable, and therefore, after a suitable change of bases, we can further assume that

$$A = \text{diag}(a_1, \dots, a_L) \quad \text{with} \quad \{a_1, \dots, a_L\} = \sigma(A)$$

and

$$a_1 \geq \dots \geq a_L > 0. \tag{2}$$

Notice that $\tau_A = 1$ implies that A is unitary and the above will imply that $A = I$, the identity (this is our previous case). Then the strict grout condition seems to be necessary always (apart from the case of our previous articles).

From now on, for the rest of the paper, we will assume that A is of this form.

2 Generalized eigenfunctions of the free Hamiltonian

We start with the definition of a generalized eigenfunction:

Definition 2.1. Let $N \in \mathbb{N}$. For a matrix-valued function $\phi : \mathbb{Z} \rightarrow \mathcal{M}_{L \times N}(\mathbb{C})$, by abuse of notation, we define

$$(H_0\Phi)(n) := A\Phi(n+1) + A\Phi(n-1)$$

and

$$(V\Phi)(n) := V(n)\Phi(n)$$

and set

$$H := H_0 + V.$$

Further, given $K \in \{H_0, H\}$ and an energy $E \in \mathbb{C}$, we write $\mathcal{G}_N(K, E)$ for the set of generalized eigenfunctions of E to the generalized eigenvalue E , i.e. for the set of functions $\phi : \mathbb{Z} \rightarrow \mathcal{M}_{L \times N}(\mathbb{C})$, such that

$$K\phi = E\phi.$$

We write

$$\mathcal{G}(K, E) := \mathcal{G}_1(K, E)$$

for the space of generalized eigenfunctions with values in $\mathbb{C}^L \cong \mathcal{M}_{L \times 1}(\mathbb{C})$.

Remark 2.2. Since a solution $\phi \in \mathcal{G}_N(K, E)$ is uniquely determined by $\phi(0)$ and $\phi(1)$, one has

$$\dim \mathcal{G}_N(K, E) = 2LN,$$

so especially

$$\dim \mathcal{G}(K, E) = 2L.$$

Further, given a function $\phi \in \mathcal{G}_N(K, E)$, for every $v \in \mathbb{C}^N$, one has $\phi \cdot v \in \mathcal{G}(K, E)$.

To construct some explicit generalized eigenfunctions, for a matrix $Z \in \text{GL}_L(\mathbb{C})$, we consider the function $\phi : \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C})$ defined by

$$\phi(n) := Z^n.$$

We then have

$$H_0\phi = A(Z + Z^{-1})\phi.$$

Therefore, one has $\phi \in \mathcal{G}_L(H_0, E)$, if and only if

$$A(Z + Z^{-1}) = E\mathbf{1}_L.$$

Formally, the equation

$$z + z^{-1} = w$$

has the solutions

$$z = \frac{w \pm \sqrt{w^2 - 4}}{2} = \frac{w \pm \sqrt{w-2}\sqrt{w+2}}{2}.$$

Since we are dealing with complex numbers we have to be carefull about the choice of the branch of the square root we take. This motivates the following definition:

Definition 2.3. For $\sigma \in \{\pm\}$ and a subset $M \subseteq \mathbb{R}$, we set

$$\mathbb{C}_M^\sigma := \mathbb{C} \setminus \{z \in \mathbb{C} : \text{Re}(z) \in M \text{ and } \text{Im}(z) \in \sigma i(-\infty, 0)\}.$$

We consider the unique continuous square root

$$\sqrt{\cdot}^\sigma : \mathbb{C}_{\{0\}}^\sigma = \mathbb{C} \setminus \sigma i(-\infty, 0) \rightarrow \mathbb{C}$$

that is holomorphic on $\mathbb{C} \setminus \sigma i(-\infty, 0]$ and satisfies

$$\sqrt{[0, \infty)}^\sigma = [0, \infty).$$

Further, we define the function

$$z_\sigma : \mathbb{C}_{\{-2, 2\}}^\sigma \rightarrow \mathbb{C}, \quad w \mapsto \frac{w - \sqrt{w-2}^\sigma \sqrt{w+2}^\sigma}{2}.$$

Remark 2.4. Since, for $z \in [0, \infty)$, one has

$$\overline{\sqrt{z}^{-\sigma}} = \sqrt{z}^{\sigma}$$

and both functions

$$\mathbb{C} \setminus \sigma i(-\infty, 0] \rightarrow \mathbb{C}, \quad z \mapsto \sqrt{z}^{\sigma}$$

and

$$\mathbb{C} \setminus \sigma i(-\infty, 0] \rightarrow \mathbb{C}, \quad z \mapsto \overline{\sqrt{z}^{-\sigma}}$$

are holomorphic, one has

$$\overline{\sqrt{z}^{-\sigma}} = \sqrt{z}^{\sigma} \quad \forall z \in \mathbb{C} \setminus -\sigma i(0, \infty).$$

Lemma 2.5. *Let $\sigma \in \{\pm 1\}$ and $w \in \mathbb{C}_{\{-2, 2\}}^{\sigma}$. Then the following hold:*

- (a) $z_{\sigma}(w) \neq 0$ and $z_{\sigma}(w) + z_{\sigma}(w)^{-1} = w$.
- (b) One has $|z_{\sigma}(w)| = 1$, if and only if $w \in [-2, 2]$.
- (c) If $|\operatorname{Re}(w)| > 2$ or $\sigma \operatorname{Im}(w) > 0$, one has $|z_{\sigma}(w)| < 1$.
- (d) If $|\operatorname{Re}(w)| < 2$ and $\sigma \operatorname{Im}(w) < 0$, one has $|z_{\sigma}(w)| > 1$.
- (e) If $\sigma \operatorname{Im}(w) > 0$, one has $0 < -\sigma \operatorname{Im} z_{\sigma}(w)$.
- (f) If $\sigma \operatorname{Im}(w) \geq 0$, one has $0 \leq -\sigma \operatorname{Im} z_{\sigma}(w)$.
- (g) One has $z_{\sigma}(w) = \pm 1$, if and only if $w = \pm 2$.
- (h) One has $z_{\sigma}(w) = \overline{z_{-\sigma}(\overline{w})}$.
- (i) If $|\operatorname{Re}(w)| > 2$, one has $z_{\sigma}(w) = z_{-\sigma}(w)$.
- (j) If $|\operatorname{Re}(w)| < 2$, one has $z_{\sigma}(w) = z_{-\sigma}(w)^{-1}$.
- (k) If $|\operatorname{Re}(w)| > 2$ or $\sigma \operatorname{Im}(w) > 0$, one has $|z_{\sigma}(w)| \leq \frac{3}{|w|+1}$.

Proof. (a) One has

$$\begin{aligned} z_{\sigma}(w) \cdot \frac{w + \sqrt{w-2}^{\sigma} \sqrt{w+2}^{\sigma}}{2} &= \frac{w - \sqrt{w-2}^{\sigma} \sqrt{w+2}^{\sigma}}{2} \cdot \frac{w + \sqrt{w-2}^{\sigma} \sqrt{w+2}^{\sigma}}{2} \\ &= \frac{w^2 - \left(\sqrt{w-2}^{\sigma} \sqrt{w+2}^{\sigma}\right)^2}{4} \\ &= \frac{w^2 - (w-2)(w+2)}{4} = \frac{w^2 - (w^2 - 4)}{4} = 1, \end{aligned}$$

so $z_{\sigma}(w) \neq 0$ and

$$z_{\sigma}(w)^{-1} = \frac{w + \sqrt{w-2}^{\sigma} \sqrt{w+2}^{\sigma}}{2}. \quad (3)$$

This implies

$$z_{\sigma}(w) + z_{\sigma}(w)^{-1} = \frac{w - \sqrt{w-2}^{\sigma} \sqrt{w+2}^{\sigma}}{2} + \frac{w + \sqrt{w-2}^{\sigma} \sqrt{w+2}^{\sigma}}{2} = w.$$

(b) For $w \in [-2, 2]$, one has $2 \pm w \geq 0$ and therefore

$$z_\sigma(w) = \frac{w - \sqrt{w-2}^\sigma \sqrt{w+2}^\sigma}{2} = \frac{w - \sigma i \sqrt{2-w} \sqrt{2+w}}{2} = \frac{w - \sigma i \sqrt{4-w^2}}{2},$$

so

$$|z_\sigma(w)|^2 = \frac{w^2 + \sqrt{4-w^2}^2}{4} = 1.$$

If, on the other hand $|z_\sigma(w)| = 1$, by (a), one has

$$w = z_\sigma(w) + z_\sigma(w)^{-1} = z_\sigma(w) + \overline{z_\sigma(w)} = 2 \operatorname{Re}(z_\sigma(w)) \in [-2, 2].$$

(c) Since the set

$$M := \{w \in \mathbb{C} : |\operatorname{Re}(w)| > 2 \text{ or } \sigma \operatorname{Im}(w) > 0\}$$

has trivial intersection with the interval $[-2, 2]$, by (b), for every $w \in M$, one either has $|z_\sigma(w)| < 1$ or $|z_\sigma(w)| > 1$. If there would be $w, w' \in M$ with $|z_\sigma(w)| < 1$ and $|z_\sigma(w')| > 1$, since M is path connected, by the intermediate value theorem there would exist $\tilde{w} \in M$ with $|z_\sigma(\tilde{w})| = 1$, which we already showed can not occur. It is therefore sufficient to show that $|z_\sigma(w)| < 1$ for one $w \in M$. We have

$$|z_\sigma(3)| = \left| \frac{3 - \sqrt{3-2}^\sigma \sqrt{3+2}^\sigma}{2} \right| = \left| \frac{3 - \sqrt{5}}{2} \right| < 1.$$

(d) Using that the set

$$N := \{w \in \mathbb{C} : |\operatorname{Re}(w)| < 2 \text{ and } \sigma \operatorname{Im}(w) < 0\}$$

is path connected and has trivial intersection with the interval $[-2, 2]$, as in (c), one argues that either $|z_\sigma(w)| > 1$ for every $w \in N$ or $|z_\sigma(w)| < 1$ for every $w \in N$. In the latter case, by (b) and (c), we would have $|z_\sigma(w)| \leq 1$ for every $w \in \mathbb{C}_{\{-2,2\}}^\sigma$, which, by (b) and the maximum modulus principle, would imply that the function z_σ is constant, in contradiction to (a).

(e) By (a), we have

$$\begin{aligned} 0 &< \sigma \operatorname{Im}(w) = \sigma \operatorname{Im}(z_\sigma(w) + z_\sigma(w)^{-1}) \\ &= \sigma \operatorname{Im}\left(z_\sigma(w) + \overline{z_\sigma(w)} |z_\sigma(w)|^{-2}\right) = \sigma (1 - |z_\sigma(w)|^{-2}) \cdot \operatorname{Im}(z_\sigma(w)). \end{aligned}$$

Using that, by (c), one has $1 - |z_\sigma(w)|^{-2} < 0$, this implies

$$0 < -\sigma \operatorname{Im}(z_\sigma(w)).$$

(f) This follows immediately by (e) and the continuity of the function z_σ .

(g) This follows immediately by (a).

(h) This follows immediately by Remark 2.4.

(i) This follows from the fact that, for $\operatorname{Re}(w) > 2$, one has

$$\sqrt{w-2}^\sigma \sqrt{w+2}^\sigma = \sqrt{w-2}^{-\sigma} \sqrt{w+2}^{-\sigma}$$

and for $\operatorname{Re}(w) < -2$, one has

$$\sqrt{w-2}^\sigma \sqrt{w+2}^\sigma = \left(-\sqrt{w-2}^{-\sigma}\right) \left(-\sqrt{w+2}^{-\sigma}\right) = \sqrt{w-2}^{-\sigma} \sqrt{w+2}^{-\sigma}.$$

(j) This follows from Equation (3) and the fact that, for $|\operatorname{Re}(w)| < 2$, one has

$$\sqrt{w-2}^\sigma \sqrt{w+2}^\sigma = \left(-\sqrt{w-2}^{-\sigma}\right) \sqrt{w+2}^{-\sigma} = -\sqrt{w-2}^{-\sigma} \sqrt{w+2}^{-\sigma}.$$

(k) By (a) and (c), we have

$$|w| = |z_\sigma(w) + z_\sigma(w)^{-1}| \leq |z_\sigma(w)| + |z_\sigma(w)|^{-1} \leq 1 + |z_\sigma(w)|^{-1}.$$

For $|w| > 1$, this yields

$$|z_\sigma(w)| \leq \frac{1}{|w| - 1}.$$

Therefore, for $|w| \geq 2$, we have

$$|z_\sigma(w)| \leq \frac{1}{|w| - 1} = \frac{3}{|w| + (2|w| - 3)} \leq \frac{3}{|w| + 1}.$$

On the other hand, by (c), for $|w| \leq 2$, we have

$$|z_\sigma(w)| \leq 1 = \frac{3}{2 + 1} \leq \frac{3}{|w| + 1}.$$

□

Definition 2.6. We set

$$\mathfrak{a} := \sigma(2A) \cup \sigma(-2A) = \{2a_1, \dots, 2a_L\} \cup \{-2a_1, \dots, -2a_L\}.$$

Then, for $\sigma \in \{\pm\}$, we define

$$Z_\sigma : \mathbb{C}_\mathfrak{a}^\sigma \rightarrow \mathcal{M}_L(\mathbb{C}), \quad E \mapsto \operatorname{diag} \left(z_\sigma \left(\frac{E}{a_1} \right), \dots, z_\sigma \left(\frac{E}{a_L} \right) \right).$$

Proposition 2.7. *Let $\sigma \in \{\pm\}$ and $E \in \mathbb{C}_\mathfrak{a}^\sigma$. Then the following hold:*

(a) *One has $Z_\sigma(E) \in \operatorname{GL}_L(\mathbb{C})$ and*

$$A(Z_\sigma(E) + Z_\sigma(E)^{-1}) = E\mathbf{1}_L.$$

(b) *If $|\operatorname{Re}(E)| > 2a_1$ or $\sigma \operatorname{Im}(E) \geq 0$, one has*

$$\|Z_\sigma(E)\| \leq 1$$

with strict inequality holding if $|\operatorname{Re}(E)| > 2a_1$ or $\sigma \operatorname{Im}(E) > 0$.

(c) One has

$$Z_\sigma(E) = Z_{-\sigma}(\overline{E})^*.$$

(d) One has $Z_\sigma(E) - Z_\sigma(E)^{-1} \notin \text{GL}_L(\mathbb{C})$, if and only if $E \in \mathfrak{a}$.

(e) For $k \in \{1, \dots, L\}$ and $|\text{Re}(E)| > 2a_k$, one has

$$Z_\sigma(E)e_k = Z_{-\sigma}(E)e_k.$$

(f) For $k \in \{1, \dots, L\}$ and $|\text{Re}(E)| < 2a_k$, one has

$$Z_\sigma(E)e_k = Z_{-\sigma}(E)^{-1}e_k.$$

(g) If $E \in \mathbb{C}_{[-2a_1, 2a_1] \setminus (-2a_L, 2a_L)}^\sigma$, one has

$$\|Z_\sigma(E)\| \cdot \|Z_\sigma(E)^{-1}\| \leq \tau_A,$$

with equality holding, if and only if $E \in \{-2a_1, 2a_1\}$ or $\tau_A = 1$.

(h) If $|\text{Re}(E)| > 2a_1$ or $\sigma \text{Im}(E) \geq 0$, one has

$$\|Z_\sigma(E)\| \leq \frac{3}{|E| \|A\|^{-1} + 1}.$$

Proof. (a) By Lemma 2.5(a), we immediately get $Z_\sigma(E) \in \text{GL}_L(\mathbb{C})$ and

$$Z_\sigma(E) + Z_\sigma(E)^{-1} = EA^{-1}.$$

(b) This follows immediately by Lemma 2.5(b),(c).

(c) This follows immediately by Lemma 2.5(g).

(d) One has $Z_\sigma(E) - Z_\sigma(E)^{-1} \notin \text{GL}_L(\mathbb{C})$, if and only if there exists $k = 1, \dots, L$ such that

$$z_\sigma\left(\frac{E}{a_k}\right) = z_\sigma\left(\frac{E}{a_k}\right)^{-1}.$$

This is equivalent to

$$z_\sigma\left(\frac{E}{a_k}\right) = \pm 1,$$

which, by Lemma 2.5(h), is equivalent to

$$\frac{E}{a_k} = \pm 2, \quad \text{i.e.} \quad E = \pm 2a_k.$$

(e) This follows immediately by Lemma 2.5(i).

(f) This follows immediately by Lemma 2.5(j).

(g) The proof of this statement is quite lengthy and not very insightful, which is why we decided to move it to Appendix A.

(h) By Lemma 2.5(k), we immediately get

$$\|Z_\sigma(E)\| = \max_{k=1, \dots, L} \left| z_\sigma\left(\frac{E}{a_k}\right) \right| \leq \max_{k=1, \dots, L} \frac{3}{\left|\frac{E}{a_k}\right| + 1} = \frac{3}{\left|\frac{E}{a_1}\right| + 1} = \frac{3}{|E| \|A\|^{-1} + 1}. \quad \square$$

From now on, for better readability, we will often omit the dependence of z_σ and Z_σ on E . We can now construct some explicit general eigenfunctions:

Definition 2.8. For $E \in \mathbb{C}$, we define

$$I^E := \text{span} \left\{ k \in \{1, \dots, L\} : \frac{E}{a_k} \in \{-2, 2\} \right\}.$$

Further, we define $P_{\pm}^E \in B(\mathbb{C}^L)$ to be the orthogonal projection onto the subspace

$$\text{span} \{ e_k : k \in I^E \}.$$

Remark 2.9. For $E \in \mathbb{C}_a^\sigma$, one has $e_k \in P_{\pm}^E \mathbb{C}^L$, if and only if

$$Z_\sigma(E) e_k = z_\sigma \left(\frac{E}{a_k} \right) e_k = z_\sigma(\pm 2) e_k = (\pm 1) e_k,$$

so

$$P_{\pm}^E \mathbb{C}^L = \ker(Z_\sigma(E) - \mathbf{1}_L) \oplus \ker(Z_\sigma(E) + \mathbf{1}_L).$$

Definition 2.10. For $\sigma, \tau \in \{\pm\}$ and $E \in \mathbb{C}_a^\sigma$, we define $\phi_\tau^{E,\sigma}, \psi_\tau^{E,\sigma} : \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C})$ by

$$\phi_\tau^{E,\sigma}(n) := Z_\sigma(E)^{\tau n}$$

and

$$\psi_\tau^{E,\sigma}(n) := Z_\sigma(E)^{-\tau n} [(\mathbf{1}_L - P_{\pm}^E) + n P_{\pm}^E].$$

Further, we define $s^{E,\sigma} \in G_L(H_0, E)$ to be the unique generalized eigenfunction with $s^{E,\sigma}(0) = 0$ and $s^{E,\sigma}(1) = \mathbf{1}_L$.

Lemma 2.11. Let $\sigma, \tau \in \{\pm\}$ and $E \in \mathbb{C}_a^\sigma$. Then, the following hold:

(a) $\phi_\tau^{E,\sigma} \in \mathcal{G}_L(H_0, E)$.

(b) $\psi_\tau^{E,\sigma} \in \mathcal{G}_L(H_0, E)$.

(c) For $n \in \mathbb{N}_0$, one has $-s^{E,\sigma}(-n) = s^{E,\sigma}(n) = \sum_{k=1}^n Z_\sigma(E)^{2k-n-1} \dots$ [TODO] (make this the definitio and make the definition part of the lemma)

Proof. (a) We have

$$(H_0 \phi_\tau^{E,\sigma})(n) = A Z_\sigma^\tau(n+1) + A Z_\sigma^\tau(n-1) = A [Z_\sigma + Z_\sigma^{-1}] Z_\sigma^{\tau n} = E \phi_\tau^{E,\sigma}(n),$$

using Proposition 2.7(a) in the last step.

(b) We have

$$\begin{aligned} (H_0 \psi_\tau^{E,\sigma})(n) &= A Z_\sigma(E)^{-\tau(n+1)} [(\mathbf{1}_L - P_{\pm}^E) + (n+1) P_{\pm}^E] \\ &\quad + A Z_\sigma(E)^{-\tau(n-1)} [(\mathbf{1}_L - P_{\pm}^E) + (n-1) P_{\pm}^E] \\ &= A [Z_\sigma + Z_\sigma^{-1}] Z_\sigma(E)^{-\tau n} [(\mathbf{1}_L - P_{\pm}^E) + n P_{\pm}^E] \\ &\quad + A [Z_\sigma(E)^{-\tau(n+1)} - Z_\sigma(E)^{-\tau(n-1)}] P_{\pm}^E \\ &= E \psi_\tau^{E,\sigma}(n), \end{aligned}$$

using Proposition 2.7(a) in the last step and the fact that

$$[Z_\sigma(E)^{-\tau(n+1)} - Z_\sigma(E)^{-\tau(n-1)}] P_{\pm}^E = [(\pm 1)^{-\tau(n+1)} - (\pm 1)^{-\tau(n-1)}] P_{\pm}^E = 0.$$

(c) We define $\phi : \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C})$ by

$$\phi(n) = \begin{cases} \sum_{k=1}^n Z_\sigma^{2k-n-1} & \text{if } n \in \mathbb{N}_0 \\ -\phi(-n) & \text{if } n \in -\mathbb{N}. \end{cases}$$

Then

$$\phi(0) = 0 \quad \text{and} \quad \phi(1) = Z_\sigma^{2 \cdot 1 - 1 - 1} = Z_\sigma^0 = \mathbf{1}_L.$$

It therefore remains to show that $\phi \in \mathcal{G}(H_0, E)$. For $n = 0$, we get

$$(H_0\phi)(n) = A\phi(1) + A\phi(-1) = A\phi(1) - A\phi(1) = 0 = E\phi(0).$$

For $n \in \mathbb{N}$, one has

$$\begin{aligned} (H_0\phi)(n) &= A \sum_{k=1}^{n-1} Z_\sigma^{2k-(n-1)-1} + A \sum_{k=1}^{n+1} Z_\sigma^{2k-(n+1)-1} \\ &= A \sum_{k=1}^n Z_\sigma^{2k-(n-1)-1} + A \sum_{k=1}^n Z_\sigma^{2k-(n+1)-1} \\ &= A [Z_\sigma + Z_\sigma^{-1}] \sum_{k=1}^n Z_\sigma^{2k-n-1} = E\phi(n), \end{aligned}$$

using Proposition 2.7(a) in the last step. Finally, for $n \in -\mathbb{N}$, this yields

$$\begin{aligned} (H_0\phi)(n) &= -A\phi(-(n-1)) - A\phi(-(n+1)) = -A\phi(-n+1) - A\phi(-n-1) \\ &= -(H_0\phi)(-n) = -E\phi(-n) = E\phi(n). \end{aligned} \quad \square$$

We now show that the above-constructed functions $\phi_\tau^{E,\sigma}, \psi_\tau^{E,\sigma} \in \mathcal{G}_L(H_0, E)$ give rise to all generalized eigenfunctions and show some of their basic properties:

Lemma 2.12. *Let $\sigma, \tau \in \{\pm\}$, $E \in \mathbb{C}_a^\sigma$ and $N \in \mathbb{N}$. Then the following hold:*

(a) *One has*

$$\mathcal{G}_N(H_0, E) = \phi_\tau^{E,\sigma} \mathcal{M}_{L \times N}(\mathbb{C}) \oplus \psi_\tau^{E,\sigma} \mathcal{M}_{L \times N}(\mathbb{C}).$$

(b) *Let $E \in \mathbb{C}_{[-2a_1, 2a_1]}^\sigma$. Then, for $\phi \in \mathcal{G}_N(H_0, E)$ and $M \in \mathcal{M}_{L \times N}(\mathbb{C})$, one has*

$$\lim_{n \rightarrow \tau\infty} \phi_\tau^{E,\sigma}(n)^{-1} \phi(n) = M \quad \Leftrightarrow \quad \phi = \phi_\tau^{E,\sigma} \cdot M.$$

(c) *If $E \in \mathbb{C}_{[-2a_1, 2a_1]}^\sigma$, for every $n \in \tau\mathbb{N}_0$, one has*

$$\|\phi_\tau^{E,\sigma}(n)\| \leq 1.$$

(d) *For $k \in \{1, \dots, L\}$ and $|\operatorname{Re}(E)| > 2a_k$, one has*

$$\phi_\tau^{E,\sigma} e_k = \phi_\tau^{E,-\sigma} e_k.$$

(e) *For $k \in \{1, \dots, L\}$ and $|\operatorname{Re}(E)| < 2a_k$, one has*

$$\phi_\tau^{E,\sigma} e_k = \phi_{-\tau}^{E,-\sigma} e_k.$$

Proof. (a) The inclusion

$$\phi_\tau^{E,\sigma} \mathcal{M}_{L \times N}(\mathbb{C}) + \psi_\tau^{E,\sigma} \mathcal{M}_{L \times N}(\mathbb{C}) \subseteq \mathcal{G}_N(H_0, E)$$

follows immediately by Lemma 2.11(a). Since, by Remark 2.2, we have

$$\dim \mathcal{G}_N(H_0, E) = 2LN,$$

it suffices to show that the map

$$\mathcal{M}_{L,N}(\mathbb{C})^2 \ni (B, C) \mapsto \phi_\tau^{E,\sigma} B + \psi_\tau^{E,\sigma} C$$

is injective. So let

$$0 = \phi_\tau^{E,\sigma} B + \psi_\tau^{E,\sigma} C.$$

Then

$$0 = \phi_\tau^{E,\sigma}(0)B + \psi_\tau^{E,\sigma}(0)C = B + (\mathbf{1}_L - P_\pm^E)C$$

and

$$0 = \phi_\tau^{E,\sigma}(1)B + \psi_\tau^{E,\sigma}(1)C = Z_\sigma^{\tau 1} B + Z_\sigma^{-\tau 1} C.$$

Multiplying the first equality from the left with $P_\pm^E$ and $1 - P_\pm^E$ respectively, yields

$$P_\pm^E B = 0 \quad \text{and} \quad (\mathbf{1} - P_\pm^E)B = -(\mathbf{1}_L - P_\pm^E)C.$$

Multiplying the second equality from the left with $1 - P_\pm^E$ and using that $[Z_\sigma, P_\pm^E] = 0$, we get

$$\begin{aligned} 0 &= Z_\sigma^{\tau 1} (\mathbf{1}_L - P_\pm^E)B + Z_\sigma^{-\tau 1} (\mathbf{1}_L - P_\pm^E)C \\ &= Z_\sigma^{\tau 1} (\mathbf{1}_L - P_\pm^E)B - Z_\sigma^{-\tau 1} (\mathbf{1}_L - P_\pm^E)B = \tau [Z_\sigma - Z_\sigma^{-1}] (\mathbf{1}_L - P_\pm^E)B. \end{aligned}$$

Since, by ...[TODO], we have $\ker(Z_\sigma - Z_\sigma^{-1}) = P_\pm^E \mathbb{C}^L$, this yields

$$(\mathbf{1}_L - P_\pm^E)B = P_\pm^E (\mathbf{1}_L - P_\pm^E)B = 0.$$

We therefore have $B = 0$ and therefore also

$$C = -Z_\sigma^{\tau 2} B = 0.$$

(b) Assume that

$$\lim_{n \rightarrow \tau\infty} \phi_\tau^{E,\sigma}(n)^{-1} \phi(n) = M.$$

By (a), there exist matrices $M_\phi, M_\psi \in \mathcal{M}_{L \times N}(\mathbb{C})$, such that

$$\phi = \phi_\tau^{E,\sigma} M_\phi + \psi_\tau^{E,\sigma} M_\psi.$$

Then

$$\begin{aligned} M &= \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,\sigma}(n)^{-1} \phi(n) = \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,\sigma}(n)^{-1} [\phi_\tau^{E,\sigma}(n) M_\phi + \psi_\tau^{E,\sigma}(n) M_\psi] \\ &= M_\tau + \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,\sigma}(n)^{-1} \psi_\tau^{E,\sigma}(n) M_\psi \\ &= M_\tau + \lim_{n \rightarrow \tau\infty} Z_\sigma(E)^{-\tau 2n} [(\mathbf{1}_L - P_\pm^E) + n P_\pm^E] M_\psi \\ &= M_\tau + \lim_{n \rightarrow \tau\infty} \sum_{k \in \{1, \dots, L\} \setminus I^E} |e_k\rangle z_\sigma \left(\frac{E}{a_k} \right)^{-\tau 2n} \langle e_k | M_\psi + n P_\pm^E M_\psi \end{aligned} \tag{4}$$

Since the vectors $|e_k\rangle$, $k = 1, \dots, L$ are linearly independent, the existence of the latter limit implies that $P_{\underline{}}^E M_\psi = 0$ and that, for every $k \in \{1, \dots, L\} \setminus I^E$, one of the following holds:

- (1) $\left| z_\sigma \left(\frac{E}{a_k} \right) \right| > 1$
- (2) $z_\sigma \left(\frac{E}{a_k} \right) = \pm 1$
- (3) $\langle e_k | M_\psi = 0$

Since (1) contradicts Lemma 2.5(b),(c) and (2) contradicts Lemma 2.5(g), we have $\langle e_k | M_\psi = 0$ for every $k \in \{1, \dots, L\} \setminus I^E$, i.e. $(\mathbf{1}_L - P_{\underline{}}^E) M_\psi = 0$. This yields $M_\psi = 0$. With Equation (4), this implies $M_\tau = M$ and therefore $\phi = \phi_\tau^{E,\sigma}(n)M$. The converse implication is trivial.

(c) By Proposition 2.7(b), we have

$$\|\phi_\tau^{E,\sigma}(n)\| = \|Z_\sigma^{\tau n}\| \leq \|Z_\sigma\|^{\tau n} \leq 1.$$

(d) This follows immediately by Proposition 2.7(e).

(e) This follows immediately by Proposition 2.7(f). $\square$

3 Existence of Jost solutions

After considering generalized eigenfunctions of H_0 , we now show – based on the proof in [Te00, Lem. 7.8] and [BFGS22, Thm. 33] *Corregir* – the existence of Jost solutions, i.e. given $\phi \in \mathcal{G}_L(H_0, E)$ and $\tau \in \{\pm\}$, we want to find $u \in \mathcal{G}_L(H_0, E)$ such that

$$\lim_{n \rightarrow \tau\infty} \phi(n)^{-1} u(n) = \mathbf{1}_L.$$

To avoid complicated notation, we will just execute the proof of existence of such Jost solutions for the case $\tau = +$, the case $\tau = -$ can be shown exactly analogously. We will prove the existence by solving so-called Volterra equations, i.e. equations of the form

$$u(n) = \phi(n) + \sum_{j=N}^{\infty} k(n, j) u(j)$$

for some kernel $k : \mathbb{Z}^2 \rightarrow \mathcal{M}_L(\mathbb{C})$. The following lemma shows that this in fact produces functions $u \in \mathcal{G}_L(H, E)$:

Lemma 3.1. *Let $E \in \mathbb{C}$, $\phi \in \mathcal{G}_L(H_0, E)$. and $K : \mathbb{Z}^2 \rightarrow \mathcal{M}_L(\mathbb{C})$, such that, for every $j \in \mathbb{Z}$, one has*

$$(H_0 - E)K(\cdot, j) = \delta_j \cdot \mathbf{1}_L.$$

Then, given $N \in \mathbb{Z}$, for every function $u : \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C})$ with

$$\sum_{j=N}^{\infty} \|K(n, j) u(j)\| < \infty \quad \forall n \geq N \tag{5}$$

satisfying the Volterra equation

$$u(n) = \phi(n) - \sum_{j=N}^{\infty} K(n, j)V(j)u(j) \quad \forall n \geq N,$$

one has

$$(H - E)u(n) = 0 \quad \forall n > N.$$

Proof. Since all appearing sums converge absolutely by assumption (5), we get

$$\begin{aligned} (H - E)u(n) &= V(n)u(n) + (H_0 - E)u(n) \\ &= V(n)u(n) + (H_0 - E)\phi(n) - \sum_{j=N}^{\infty} (H_0 - E)K(\cdot, j)(n)V(j)u(j) \\ &= V(n)u(n) + 0 - \sum_{j=N}^{\infty} \delta_j(n) \cdot \mathbf{1}_L \cdot V(j)u(j) = 0. \end{aligned} \quad \square$$

The above lemma motivates the following definition:

Definition 3.2. Let $\Omega \subseteq \mathbb{C}$ and consider maps

$$E : \Omega \rightarrow \mathbb{C}, \quad K : \Omega \times \mathbb{Z} \times \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C}) \quad \text{and} \quad \phi : \Omega \times \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C}).$$

We say that the quadruple (Ω, E, K, ϕ) is *admissible*, if the following hold:

(a) For every $z \in \Omega$ and $j \in \mathbb{Z}$, one has

$$(H_0 - E(z))K(z, \cdot, j) = \delta_j \cdot \mathbf{1}_L.$$

(b) For every $z \in \Omega$, one has $\phi(z, \cdot) \in \mathcal{G}_L(H_0, E(z))$, i.e.

$$(H_0 - E(z))\phi(z, \cdot) = 0.$$

(c) There exists $N \in \mathbb{Z}$ such that, for every $z \in \Omega$ and $n \geq N$, one has $\phi(z, n) \in \text{GL}_L(\mathbb{C})$ and

$$\sum_{j=N}^{\infty} \sup_{\substack{z \in \Omega \\ n \geq N}} \|\phi(z, n)^{-1} K(z, n, j)V(j)\phi(z, j)\| < \infty.$$

and

$$\lim_{n \rightarrow \infty} \sum_{j=N}^{\infty} \sup_{z \in \Omega} \|\phi(z, n)^{-1} K(z, n, j)V(j)\phi(z, j)\| = 0.$$

We have the following theorem:

Theorem 3.3. Let (Ω, E, K, ϕ) be an admissible quadruple. Then, there exist $N \in \mathbb{Z}$ and a unique function

$$u : \Omega \times \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C})$$

such that the following hold:

(a) *One has*

$$u(z, n) = \phi(z, n) - \sum_{j=N}^{\infty} K(z, n, j)V(j)u(z, j) \quad \forall n \geq N, z \in \Omega.$$

(b) *One has*

$$\lim_{n \rightarrow \infty} \sup_{z \in \Omega} \|\phi(z, n)^{-1}u(z, n) - \mathbf{1}_L\| = 0.$$

(c) *For every $z \in \Omega$, one has*

$$u(z, \cdot) \in \mathcal{G}(H, E(z)).$$

Further, if, for every $n, j \in \mathbb{Z}$ the maps

$$\Omega \ni z \mapsto \phi(z, n) \quad \text{and} \quad \Omega \ni z \mapsto K(z, n, j)$$

are continuous/holomorphic, then also the maps

$$\Omega \ni z \mapsto u(z, n), \quad n \in \mathbb{Z},$$

are continuous/holomorphic.

Proof. By assumption, there exists $N \in \mathbb{Z}$ such that, for every $z \in \Omega$ and $n \geq N$, one has $\phi(z, n) \in \text{GL}_L(\mathbb{C})$ and

$$\sum_{j=N}^{\infty} \sup_{\substack{z \in \Omega \\ n \geq N}} \|\phi(z, n)^{-1}K(z, n, j)V(j)\phi(z, j)\| < \infty.$$

By increasing N , we can assume that

$$\sum_{j=N}^{\infty} \sup_{\substack{z \in \Omega \\ n \geq N}} \|\phi(z, n)^{-1}K(z, n, j)V(j)\phi(z, j)\| \leq \frac{1}{2}.$$

Now, given $z \in \Omega$, we define an operator T^z on $\ell^\infty(\mathbb{N} \cap [N, \infty), \mathcal{M}_L(\mathbb{C}))$ by

$$(T^z f)(n) := - \sum_{j=N}^{\infty} \phi(z, n)^{-1}K(z, n, j)V(j)\phi(z, j)f(j). \quad (6)$$

Then

$$\|T^z\| \leq \sup_{n \geq N} \sum_{j=N}^{\infty} \|\phi(z, n)^{-1}K(z, n, j)V(j)\phi(z, j)\| \leq \frac{1}{2}, \quad (7)$$

so the operator $\mathbf{1} - T^z$ is invertible. We therefore can define

$$\tilde{u}^z := (\mathbf{1} - T^z)^{-1}\mathbf{1} = \sum_{l=0}^{\infty} (T^z)^l \mathbf{1} \in \ell^\infty(\mathbb{N} \cap [N, \infty), \mathcal{M}_L(\mathbb{C})), \quad (8)$$

denoting by $\mathbb{1} \in \ell^\infty(\mathbb{N} \cap [N, \infty), \mathcal{M}_L(\mathbb{C}))$ the function whose value is constantly the identity, i.e.

$$\mathbb{1}(n) = \mathbf{1}_L \quad \forall n \in \mathbb{N} \cap [N, \infty).$$

We now define $u : \Omega \times \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C})$ by

$$u(z, n) := \phi(z, n) \tilde{u}^z(n)$$

for $n \geq N$ and inductively setting

$$u(z, n) := -u(z, n+2) - A^{-1}(V(n+1) - E)u(z, n+1) \quad (9)$$

for $n < N$. We now show that the so-defined function u satisfies the desired properties.

(a) For $n \in \mathbb{N} \cap [N, \infty)$, one has

$$\mathbf{1}_L = \mathbb{1}(n) = ((\mathbf{1} - T^z) \tilde{u}^z)(n) = \tilde{u}^z(n) + \sum_{j=N}^{\infty} \phi(z, n)^{-1} K(z, n, j) V(j) \phi(z, j) \tilde{u}^z(j),$$

so $\tilde{u}^z$ satisfies the Volterra equation

$$\tilde{u}^z(n) = \mathbf{1}_L - \sum_{j=N}^{\infty} \phi(z, n)^{-1} K(z, n, j) V(j) \phi(z, j) \tilde{u}^z(j) \quad (10)$$

for $n \in \mathbb{N} \cap [N, \infty)$. Multiplying with $\phi(z, n)$ from the left yields

$$u(z, n) = \phi(z, n) - \sum_{j=N}^{\infty} K(z, n, j) V(j) u(z, j) \quad \forall n \geq N, z \in \Omega.$$

(b) By (8) and (7), for $z \in \Omega$, one has

$$\|\tilde{u}^z\|_\infty \leq \sum_{l=0}^{\infty} \|T^z\|^l \|\mathbb{1}\|_\infty \leq \sum_{l=0}^{\infty} \frac{1}{2} = 2. \quad (11)$$

Then, by Equation (10), we get

$$\begin{aligned} \sup_{z \in \Omega} \|\phi(z, n)^{-1} u(z, n) - \mathbf{1}_L\| &= \sup_{z \in \Omega} \|\tilde{u}^z(n) - \mathbf{1}_L\| \\ &= \sup_{z \in \Omega} \left\| \sum_{j=N}^{\infty} \phi(z, n)^{-1} K(z, n, j) V(j) \phi(z, j) \tilde{u}^z(j) \right\| \\ &\leq 2 \sum_{j=N}^{\infty} \sup_{z \in \Omega} \|\phi(z, n)^{-1} K(z, n, j) V(j) \phi(z, j)\| \xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

(c) For $z \in \Omega$ and $n \geq N$, by (11), we have

$$\begin{aligned} &\sum_{j=N}^{\infty} \|K(z, n, j) V(j) u(z, j)\| \\ &\leq \|\phi(z, n)\| \sum_{j=N}^{\infty} \|\phi(z, n)^{-1} K(z, n, j) V(j) \phi(z, j)\| \cdot \|\tilde{u}^z\|_\infty < \infty. \end{aligned}$$

By Lemma 3.1 and (a), this implies

$$(H - E)u(z, n) = 0 \quad \forall n > N.$$

For $n \leq N$, by (9), we have

$$\begin{aligned} (H - E)u(z, n) &= Au(z, n+1) + (V(n) - E)u(z, n) + Au(z, n-1) \\ &= Au(z, n+1) + (V(n) - E)u(z, n) + A[-u(z, n+1) - A^{-1}(V(n) - E)u(z, n)] \\ &= 0. \end{aligned}$$

We now prove the uniqueness of the function u . So let $v : \Omega \times \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C})$ be another function satisfying properties (a), (b) and (c). Given $z \in \Omega$, we define

$$\tilde{v}^z : \mathbb{Z} \cap [N, \infty) \rightarrow \mathcal{M}_L(\mathbb{C})$$

by

$$\tilde{v}^z(n) := \phi(z, n)^{-1}v(z, n).$$

Then, by (b), one has $\tilde{v}^z \in \ell^\infty(\mathbb{N} \cap [N, \infty), \mathcal{M}_L(\mathbb{C}))$ and by (a), we get

$$\tilde{v}^z = \mathbb{1} + T^z \tilde{v}^z.$$

This implies

$$\tilde{v}^z = (\mathbf{1} - T^z)^{-1} \mathbb{1} = \tilde{u}^z$$

and therefore

$$v(z, n) = \phi(z, n)\tilde{v}^z(n) = \phi(z, n)\tilde{u}^z(n) = u(z, n) \quad \forall n \geq N.$$

Since, by (c), both $u(z, \cdot), v(z, \cdot) \in \mathcal{G}(H, E(z))$ and a function in $\mathcal{G}(H, E(z))$ is uniquely determined by its values at N and $N+1$, this implies that

$$v(z, n) = u(z, n) \quad \forall n \in \mathbb{Z}.$$

If, for every $n, j \in \mathbb{Z}$ the maps

$$\Omega \ni z \mapsto \phi(z, n) \quad \text{and} \quad \Omega \ni z \mapsto K(z, n, j)$$

are continuous/holomorphic, then, due to the fact that the series in (6) converges uniformly in z , also the map

$$\Omega \ni z \mapsto T^z$$

is continuous/holomorphic. This implies that also the maps

$$\Omega \ni z \mapsto \tilde{u}^z(n) = (\mathbf{1} - T^z)^{-1} \mathbb{1}(n), \quad n \geq N$$

are continuous/holomorphic. This yields the continuity/holomorphy of the maps

$$\Omega \ni z \mapsto u(z, n) = \phi(z, n)\tilde{u}^z(n), \quad n \geq N.$$

Using Equation (9), this inductively yields the continuity/holomorphy of the maps

$$\Omega \ni z \mapsto u(z, n)$$

for $n < N$. □

We now want to use this theorem to construct explicit Jost solutions pertaining to the functions $\phi_+^{E,\sigma}$ and $\psi_+^{E,\sigma}$ constructed in Definition 2.10. For this, we will use the following kernels:

Definition 3.4. Let $\sigma \in \{\pm\}$. We define the functions

$$K_\phi^\sigma : \mathbb{C}_a^\sigma \times \mathbb{Z} \times \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C}), \quad (E, n, j) \mapsto \begin{cases} A^{-1} s^{E,\sigma}(j-n) & \text{if } j > n \\ 0 & \text{if } j \leq n \end{cases}$$

and

$$\tilde{K}_\psi^\sigma : \mathbb{C}_a^\sigma \times \mathbb{Z} \times \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C}), \quad (E, n, j) \mapsto \begin{cases} A^{-1} C^E \phi_+^{E,\sigma}(j) \psi_+^{E,\sigma}(n) & \text{if } j > n \\ A^{-1} C^E \phi_+^{E,\sigma}(n) \psi_+^{E,\sigma}(j) & \text{if } j \leq n \end{cases},$$

where

$$C^E := ([Z_\sigma - Z_\sigma^{-1}] (1 - P_\pm^E) - P_\pm^E)^{-1}.$$

Further, we set

$$K_\psi^\sigma := K_\phi^{-\sigma} P_1^E + \tilde{K}_\psi^\sigma (\mathbf{1}_L - P_1^E)$$

To construct Jost solutions using these kernels K_ϕ^σ and K_ψ^σ , we have to show that they are admissible on a suitably chosen region. These regions will be the following:

Definition 3.5. For $\sigma \in \{\pm\}$ and $E \in \mathbb{C}_a^\sigma$, we set

$$\alpha^\sigma(E) := (\max\{1, \|Z_\sigma(E)\|\})^2 \cdot \|Z_\sigma(E)\| \cdot \|Z_\sigma(E)^{-1}\|.$$

Further, for $\alpha \geq 1$, we set

$$O_\alpha^\sigma := \{E \in \mathbb{C}_a^\sigma : \alpha^\sigma(E) \leq \alpha\}.$$

Proposition 3.6. Let $\alpha \geq 1$ such that

$$\sum_{j \in \mathbb{Z}} \|V(j)\| |j| \alpha^{|j|} < \infty.$$

Then, for $\sigma \in \{\pm\}$, the following hold:

- (a) The quadruple $(O_\alpha^\sigma, \text{id}, K_\phi^\sigma, \phi_+^{\cdot,\sigma})$ is admissible.
- (b) If $\alpha > 1$ or $P_1^E \in \{0, \mathbf{1}_L\}$, the quadruple $(\{E\}, \text{id}, K_\psi^\sigma, \psi_+^{\cdot,\sigma})$ is admissible for every $E \in O_\alpha^\sigma \cap \mathbb{C}_{[-2a_1, 2a_1]}^\sigma$.

The proof of this proposition is a quite lengthy unsightful series of estimates, which is why we decided to move it to Appendix B. For now, we use it to construct Jost solutions on the sets O_α^σ and show some of their basic properties. We state the theorem for the cases $\tau \in \{\pm\}$ even though just the case $\tau = +$ is shown. The case $\tau = -$ can be shown completely analogously.

Theorem 3.7. Let $\sigma, \tau \in \{\pm\}$ and $\alpha \geq 1$ such that

$$\sum_{j \in \mathbb{Z}} \|V(j)\| |j| \alpha^{|j|} < \infty.$$

Then, for every $E \in O_\alpha^\sigma$ there exists a unique function $u_\tau^{E,\sigma} \in \mathcal{G}_L(H, E)$ such that

$$\lim_{n \rightarrow \tau_\infty} \phi_\tau^{E,\sigma}(n)^{-1} u_\tau^{E,\sigma}(n) = \mathbf{1}_L. \quad (12)$$

Further, the following hold:

(a) For every $n \in \mathbb{Z}$, the map

$$O_\alpha^\sigma \ni E \mapsto u_\tau^{E,\sigma}(n)$$

is continuous and its restriction to the interior of O_α^σ is holomorphic.

(b) For every compact subset $F \subseteq O_\alpha^\sigma \cap \mathbb{C}_{[-2a_1, 2a_1]}^\sigma$ and $N \in \mathbb{Z}$, one has

$$\sup_{E \in F, \tau n \geq N} \|u_\tau^{E,\sigma}(n)\| < \infty.$$

(c) For every $n \in \tau\mathbb{N}_0$, one has

$$\lim_{E \in O_\alpha^\sigma \cap \mathbb{C}_{[-2a_1, 2a_1]}^\sigma, |E| \rightarrow \infty} \|u_\tau^{E,\sigma}(n) - \phi_\tau^{E,\sigma}(n)\| = 0.$$

(d) For every $E \in O_\alpha^\sigma \cap \mathbb{C}_{[-2a_1, 2a_1]}^\sigma$, one has

$$\lim_{n \rightarrow \tau_\infty} u_\tau^{E,\sigma}(n) - \phi_\tau^{E,\sigma}(n) = 0.$$

Proof. (a) For every $n, j \in \mathbb{Z}$, the map

$$O_\alpha^\sigma \ni E \mapsto K^{E,\sigma}(n, j)$$

is continuous and its restriction to the interior of O_α^σ is holomorphic. This, together with the uniform convergence of the series in (6) and (8), implies that, for every $n \geq M$, the map

$$O_\alpha^\sigma \ni E \mapsto \tilde{u}_+^{E,\sigma}(n)$$

is continuous and its restriction to the interior of O_α^σ is holomorphic. Using (10), one then shows inductively that the same holds true for all $n \in \mathbb{Z}$. The statement then follows by (??) and the fact that, for every $n \in \mathbb{Z}$, the map

$$O_\alpha^\sigma \ni E \mapsto \phi_+^{E,\sigma}(n)$$

is continuous and its restriction to the interior of O_α^σ is holomorphic.

(b) We have (recall the definition of M in (??))

$$\sup_{E \in F, n \geq M} \|\tilde{u}_+^{E,\sigma}(n)\| = \sup_{E \in F} \left\| \sum_{n \in \mathbb{N}_0} (T^{E,\sigma})^n \mathbf{1} \right\|_\infty \leq \sum_{n \in \mathbb{N}_0} \left(\frac{1}{2}\right)^n = 2.$$

Together with Lemma 2.12(c), this leads

$$\sup_{E \in F, n \geq M} \|u_+^{E,\sigma}(n)\| \leq \sup_{E \in F, n \geq M} \|\phi_+^{E,\sigma}(n)\| \cdot \|\tilde{u}_+^{E,\sigma}(n)\| \leq 1 \cdot 2 = 2.$$

Now, for $N \in \mathbb{Z}$, we have

$$\sup_{E \in F, n \geq N} \|u_+^{E,\sigma}(n)\| \leq \max \left\{ \sup_{E \in F, N \leq n < M} \|u_+^{E,\sigma}(n)\|, 2 \right\} < \infty,$$

using that the set $\{n \in \mathbb{Z} : N \leq n < M\}$ is finite and that the functions

$$E \mapsto \|u_+^{E,\sigma}(n)\|$$

are continuous and therefore bounded on compact sets.

(c) By (7), we have

$$\|T^{E,\sigma}\| \leq \frac{1}{2(|E|+1)} \xrightarrow{|E| \rightarrow \infty} 0.$$

This implies (on $\ell^\infty(\mathbb{N} \cap [M, \infty), \mathcal{M}_L(\mathbb{C}))$) that

$$\tilde{u}_+^{E,\sigma} = (\mathbf{1} - T^{E,\sigma})^{-1} \mathbb{1} \xrightarrow{|E| \rightarrow \infty} \mathbb{1}.$$

The recursive relation (10) and (??) allow us to extend this formula to $\ell^\infty(\mathbb{N} \cap [N, \infty), \mathcal{M}_L(\mathbb{C}))$, for every $\mathbb{Z} \ni N < M$. Then we obtain: (please verify carefully)

$$\lim_{E \in F, |E| \rightarrow \infty} \|u_\tau^{E,\sigma}(n) - \phi_\tau^{E,\sigma}(n)\| \leq \lim_{E \in F, |E| \rightarrow \infty} \|\phi_\tau^{E,\sigma}(n)\| \cdot \|\tilde{u}_\tau^{E,\sigma}(n) - \mathbf{1}_L\| = 0,$$

using Lemma 2.12(c) in the last step.

(d) By Lemma 2.12(c) and Theorem 3.7(c), we have

$$\begin{aligned} \|u_\tau^{E,\sigma}(n) - \phi_\tau^{E,\sigma}(n)\| &\leq \|\phi_\tau^{E,\sigma}(n)\| \cdot \|\phi_\tau^{E,\sigma}(n)^{-1} u_\tau^{E,\sigma}(n) - \mathbf{1}_L\| \\ &\leq 1 \cdot \|\phi_\tau^{E,\sigma}(n)^{-1} u_\tau^{E,\sigma}(n) - \mathbf{1}_L\| \xrightarrow{n \rightarrow \tau \infty} 0. \end{aligned} \quad \square$$

Idealy, one what like the sets O_α^σ to be as big as possible to include all of the relevant energies while at the same time mainaining α relatively small, to not impose too strict growth conditions on the potential V . We will later see that both can be balanced by choosing $\alpha = \tau_A$ (...[TODO]).

We finish this section with the following analog of Lemma 2.12(d):

Lemma 3.8. *Let $\tau \in \{\pm\}$ and $\alpha \geq 1$ such that*

$$\sum_{j \in \mathbb{Z}} \|V(j)\| j \alpha^{|j|} < \infty.$$

Then, for every $E \in O_\alpha^+ \cap O_\alpha^- \cap \mathbb{R}$ and $k \in \{1, \dots, L\}$ with $|\operatorname{Re}(E)| > 2a_k$, one has

$$u_\tau^{E,+} e_k = u_\tau^{E,-} e_k.$$

Proof. By Lemma 2.12(d), for $k \in \{1, \dots, L\}$ with $|\operatorname{Re}(E)| > 2a_k$, one has

$$[\phi_\tau^{E,+}(n)^{-1} - \phi_\tau^{E,-}(n)^{-1}] e_k = [\phi_{-\tau}^{E,+}(n) - \phi_{-\tau}^{E,-}(n)] e_k = 0$$

and for $k \in \{1, \dots, L\}$ with $|\operatorname{Re}(E)| < 2a_k$, one has

$$\|[\phi_\tau^{E,+}(n)^{-1} - \phi_\tau^{E,-}(n)^{-1}] e_k\| = \left\| \left[z_+ \left(\frac{E}{a_k} \right)^{-\tau n} - z_- \left(\frac{E}{a_k} \right)^{-\tau n} \right] e_k \right\| \leq 2,$$

which yields

$$\|\phi_\tau^{E,+}(n)^{-1} - \phi_\tau^{E,-}(n)^{-1}\| \leq 2.$$

Together with Theorem 3.7(d), this implies

$$\begin{aligned} 0 &= \lim_{n \rightarrow \tau\infty} [\phi_\tau^{E,+}(n)^{-1} - \phi_\tau^{E,-}(n)^{-1}] [u_\tau^{E,-}(n) - \phi_\tau^{E,-}(n)] \\ &= \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,+}(n)^{-1} [u_\tau^{E,-}(n) - \phi_\tau^{E,-}(n)] - [\phi_\tau^{E,-}(n)^{-1} u_\tau^{E,-}(n) - \mathbf{1}_L] \\ &= \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,+}(n)^{-1} [u_\tau^{E,-}(n) - \phi_\tau^{E,-}(n)], \end{aligned}$$

using Theorem 3.7 in the last step. For $k \in \{1, \dots, L\}$ with $|\operatorname{Re}(E)| > 2a_k$, by Lemma 2.12(d), this yields

$$\begin{aligned} 0 &= \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,+}(n)^{-1} [u_\tau^{E,-}(n) - \phi_\tau^{E,-}(n)] e_k \\ &= \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,+}(n)^{-1} [u_\tau^{E,-}(n) - \phi_\tau^{E,+}(n)] e_k \\ &= \lim_{n \rightarrow \tau\infty} [\phi_\tau^{E,+}(n)^{-1} u_\tau^{E,-}(n) - \mathbf{1}_L] e_k \\ &= \lim_{n \rightarrow \tau\infty} \phi_\tau^{E,+}(n)^{-1} [u_\tau^{E,-}(n) - u_\tau^{E,+}(n)] e_k, \end{aligned}$$

using Theorem 3.7 in the last step. We therefore have

$$\lim_{n \rightarrow \tau\infty} \phi_\tau^{E,+}(n)^{-1} (u_\tau^{E,+}(n) + [u_\tau^{E,-}(n) - u_\tau^{E,+}(n)] |e_k\rangle\langle e_k|) = \mathbf{1}_L,$$

which implies

$$[u_\tau^{E,-}(n) - u_\tau^{E,+}(n)] |e_k\rangle\langle e_k| = 0$$

by the uniqueness assertion in Theorem 3.7. $\square$

4 Analytic behaviour in the thresholds

Lemma 4.1. *Let $\sigma \in \{\pm\}$ and $w \in \mathbb{C}_{\{-2,2\}}^\sigma$. Then, there exists $r > 0$ and a holomorphic function*

$$f_\sigma^w : U_r(0) \rightarrow \mathbb{C}$$

such that, for every $\zeta \in U_r(0)$, one has $f_\sigma^w(\zeta) \neq 0$ and

$$f_\sigma^w(\zeta) + f_\sigma^w(\zeta)^{-1} = \zeta^2 + w$$

and, for every $\xi \in U_{r^2}(w) \cap \mathbb{C}_{\{w\}}^\sigma$, one has

$$z_\sigma(\xi) = f_\sigma^w \left(\sqrt{\xi - w}^\sigma \right).$$

Proof. We first consider the case that $w \notin \{-2, 2\}$. Then, setting

$$r := \sqrt{\text{dist}(w, \{-2, 2\})},$$

the function z_σ is holomorphic on $U_{r^2}(w)$, so the function

$$f_\sigma^w : U_r(0) \rightarrow \mathbb{C}, \quad \zeta \mapsto z_\sigma(\zeta^2 + w)$$

is holomorphic on $U_r(0)$. By Lemma 2.5(a), for $\zeta \in U_r(0)$, we then have

$$f_\sigma^w(\zeta) = z_\sigma(\zeta^2 + w) \neq 0$$

and

$$f_\sigma^w(\zeta) + f_\sigma^w(\zeta)^{-1} = z_\sigma(\zeta^2 + w) + z_\sigma(\zeta^2 + w)^{-1} = \zeta^2 + w.$$

Further, for every $\xi \in U_{r^2}(w) \cap \mathbb{C}_{\{w\}}^\sigma$, one has

$$f_\sigma^w(\sqrt{\xi - w}^\sigma) = z_\sigma\left(\left(\sqrt{\xi - w}^\sigma\right)^2 + w\right) = z_\sigma(\xi).$$

Now, we consider the case $w \in \{-2, 2\}$. We set $r = 2$ and define

$$f_\sigma^{\pm 2} : U_r(0) \rightarrow \mathbb{C}, \quad \zeta \mapsto \frac{(\zeta^2 \pm 2) - \zeta \sqrt{\zeta^2 \pm 4}^\sigma}{2}.$$

This function is obviously holomorphic. Further, for $\zeta \in U_r(0)$, we have

$$\begin{aligned} f_\sigma^{\pm 2}(\zeta) \frac{(\zeta^2 \pm 2) + \zeta \sqrt{\zeta^2 \pm 4}^\sigma}{2} &= \frac{(\zeta^2 \pm 2) - \zeta \sqrt{\zeta^2 \pm 4}^\sigma}{2} \frac{(\zeta^2 \pm 2) + \zeta \sqrt{\zeta^2 \pm 4}^\sigma}{2} \\ &= \frac{(\zeta^2 \pm 2)^2 - \left(\zeta \sqrt{\zeta^2 \pm 4}^\sigma\right)^2}{4} \\ &= \frac{(\zeta^4 \pm 4\zeta^2 + 4) - (\zeta^4 \pm 4\zeta^2)}{4} = \frac{4}{4} = 1. \end{aligned}$$

This shows that $f_\sigma^{\pm 2}(\zeta) \neq 0$ with

$$f_\sigma^{\pm 2}(\zeta)^{-1} = \frac{(\zeta^2 \pm 2) + \zeta \sqrt{\zeta^2 \pm 4}^\sigma}{2}.$$

We therefore get

$$f_\sigma^{\pm 2}(\zeta) + f_\sigma^{\pm 2}(\zeta)^{-1} = \frac{(\zeta^2 \pm 2) - \zeta \sqrt{\zeta^2 \pm 4}^\sigma}{2} + \frac{(\zeta^2 \pm 2) + \zeta \sqrt{\zeta^2 \pm 4}^\sigma}{2} = \zeta^2 \pm 2.$$

Further, for $\xi \in U_{r^2}(\pm 2) \cap \mathbb{C}_{\{\pm 2\}}^\sigma$

$$\begin{aligned} f_\sigma^{\pm 2}(\sqrt{w \mp 2}^\sigma) &= \frac{\left(\left(\sqrt{w \mp 2}^\sigma\right)^2 \pm 2\right) - \sqrt{w \mp 2}^\sigma \sqrt{\left(\sqrt{w \mp 2}^\sigma\right)^2 \pm 4}^\sigma}{2} \\ &= \frac{w - \sqrt{w \mp 2}^\sigma \sqrt{w \pm 2}^\sigma}{2} = z_\sigma(w). \end{aligned}$$

□

Lemma 4.2. *Let $\sigma \in \{\pm\}$ and $E \in \mathbb{C}_{\mathbf{a}}^{\sigma}$. Then, there exists $r_E > 0$ and a holomorphic function*

$$F_{\sigma}^E : U_{r_E}(0) \rightarrow \mathcal{M}_L(\mathbb{C})$$

such that, for every $Y \in U_{r_E}(0)$, one has $F_{\sigma}^E(Y) \in \mathrm{GL}_L(\mathbb{C})$ and

$$A \left[F_{\sigma}^E(Y) + F_{\sigma}^E(Y)^{-1} \right] = (Y^2 + E) \mathbf{1}_L$$

and, for every $X \in U_{r_E^2}(E) \cap \mathbb{C}_{\{E\}}^{\sigma}$, one has

$$Z_{\sigma}(X) = F^{E, \sigma} \left(\sqrt{X - E}^{\sigma} \right).$$

Proof. By Lemma 4.1, there exist $r_1, \dots, r_L > 0$ and functions

$$f_{\sigma}^{\frac{E}{a_k}} : U_{r_k}(0) \rightarrow \mathbb{C}, \quad k = 1, \dots, L,$$

such that, for every $\zeta \in U_{r_k}(0)$, one has $f_{\sigma}^{\frac{E}{a_k}}(\zeta) \neq 0$ and

$$f_{\sigma}^{\frac{E}{a_k}}(\zeta) + f_{\sigma}^{\frac{E}{a_k}}(\zeta)^{-1} = \zeta^2 + \frac{E}{a_k} \quad (13)$$

and, for every $\xi \in U_{r_k^2}\left(\frac{E}{a_k}\right) \cap \mathbb{C}_{\left\{\frac{E}{a_k}\right\}}^{\sigma}$, one has

$$f_{\sigma}^{\frac{E}{a_k}} \left(\sqrt{\xi - \frac{E}{a_k}}^{\sigma} \right) = z_{\sigma}(\xi).$$

Then, for every $X \in U_{a_k r_k^2}(E) \cap \mathbb{C}_{\{-2a_k, 2a_k\}}^{\sigma}$, one has

$$f_{\sigma}^{\frac{E}{a_k}} \left(\frac{\sqrt{X - E}^{\sigma}}{\sqrt{a_k}} \right) = f_{\sigma}^{\frac{E}{a_k}} \left(\sqrt{\frac{X}{a_k} - \frac{E}{a_k}}^{\sigma} \right) = z_{\sigma} \left(\frac{X}{a_k} \right). \quad (14)$$

We now set

$$r_E := \min_{k=1, \dots, L} \sqrt{a_k} r_k > 0$$

and define

$$F_{\sigma}^E : U_{r_E}(0) \rightarrow \mathcal{M}_L(\mathbb{C}), \quad Y \mapsto \mathrm{diag} \left(f_{\sigma}^{\frac{E}{a_1}} \left(\frac{Y}{\sqrt{a_1}} \right), \dots, f_{\sigma}^{\frac{E}{a_L}} \left(\frac{Y}{\sqrt{a_L}} \right) \right).$$

Then, for $Y \in U_{r_E}(0)$, one has $F_{\sigma}^E(Y) \in \mathrm{GL}_L(\mathbb{C})$, for $k = 1, \dots, L$, by Equation (13), one has

$$a_k \left[f_{\sigma}^{\frac{E}{a_k}} \left(\frac{Y}{\sqrt{a_k}} \right) + f_{\sigma}^{\frac{E}{a_k}} \left(\frac{Y}{\sqrt{a_k}} \right)^{-1} \right] = a_k \left[\left(\frac{Y}{\sqrt{a_k}} \right)^2 + \frac{E}{a_k} \right] = Y^2 + E,$$

which yields

$$A \left[F_{\sigma}^E(Y) + F_{\sigma}^E(Y)^{-1} \right] = (Y^2 + E) \mathbf{1}_L.$$

Further, for $X \in U_{r_E^2}(E) \cap \mathbb{C}_{\{E\}}^\sigma$, by Equation (14), we have

$$\begin{aligned} F_\sigma^E(\sqrt{X-E}^\sigma) &= \text{diag} \left(f_{\sigma^{\frac{E}{a_1}}} \left(\frac{\sqrt{X-E}^\sigma}{\sqrt{a_1}} \right), \dots, f_{\sigma^{\frac{E}{a_L}}} \left(\frac{\sqrt{X-E}^\sigma}{\sqrt{a_L}} \right) \right) \\ &= \text{diag} \left(z_\sigma \left(\frac{X}{a_1} \right), \dots, z_\sigma \left(\frac{X}{a_L} \right) \right) = Z_\sigma(X). \end{aligned} \quad \square$$

Definition 4.3. Let $\sigma \in \{\pm\}$ and $E \in \mathbb{C}_a^\sigma$. Further, let $r_E > 0$ and $F_\sigma^E : U_r(0) \rightarrow \mathcal{M}_L(\mathbb{C})$ be as in Lemma 4.2. For $Y \in U_{r_E}(0)$, we define

$$\Phi_{\tau,Y}^{E,\sigma} : \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C}), \quad n \mapsto F_\sigma^E(Y)^{\tau n}$$

and

$$S_Y^{E,\sigma} : \mathbb{Z} \rightarrow \mathcal{M}_L(\mathbb{C}), \quad n \mapsto \text{sgn}(n) \sum_{k=1}^{|n|} F_\sigma^E(Y)^{2k-|n|-1}$$

and

$$K_\Phi^{E,\sigma} : U_r(0) \times \mathbb{Z} \times \mathbb{Z}, \quad (Y, n, j) \mapsto \begin{cases} A^{-1} S_Y^{E,\sigma}(j-n) & \text{if } j > n \\ 0 & \text{if } j \leq n. \end{cases}$$

Remark 4.4. By Lemma 4.2 and Lemma 2.11(c), for every $X \in U_{r_E^2}(E) \cap \mathbb{C}_a^\sigma \cap C_{\{E\}}^\sigma$, one has

$$\Phi_{\tau,\sqrt{X-E}^\sigma}^{E,\sigma} = \phi_\tau^{X,\sigma}, \quad S_{\sqrt{X-E}^\sigma}^{E,\sigma} = s^{X,\sigma} \quad \text{and} \quad K_\Phi^{E,\sigma}(\sqrt{X-E}^\sigma, \cdot, \cdot) = K_\phi^\sigma(X, \cdot, \cdot).$$

Lemma 4.5. Let $\sigma \in \{\pm\}$ and $E \in \mathbb{C}_a^\sigma$. Then, for every $Y \in U_{r_E}(0)$, one has

$$\Phi_{\tau,Y}^{E,\sigma}, S_Y^{E,\sigma} \in \mathcal{G}_L(H_0, Y^2 + E).$$

Proof. Using that, by Lemma 4.2, one has

$$F_\sigma^E(Y) + F_\sigma^E(Y)^{-1} = (Y^2 + E)\mathbf{1}_L$$

this is shown exactly as in Lemma 2.11(a),(c), replacing Z_σ with $F_\sigma^E(Y)$ and E with $(Y^2 + E)$. $\square$

Proposition 4.6. Let $\sigma \in \{\pm\}$ and $E \in \mathbb{C}_a^\sigma$. Then, the quadruple

$$(U_{r_E}(0), Y \mapsto Y^2 + E, K_\Phi^{E,\sigma}, \Phi_{\tau,\cdot}^{E,\sigma})$$

is admissible.

Proof. ...[TODO] $\square$

...[TODO]

A Norm estimates

In this section, given $w \in \mathbb{C}$ and two real numbers $a, b \in \mathbb{R}$, we want to compare the absolute values of $z_\sigma\left(\frac{w}{a}\right)$ and $z_\sigma\left(\frac{w}{b}\right)$. We start with the following lemma:

Lemma A.1. *Let $\sigma \in \{\pm\}$, $w \in \mathbb{C}_{[-2,2]}^\sigma$. Then, for every $t \in [1, \infty)$, one has $tw \in \mathbb{C}_{[-2,2]}^\sigma$ and the function*

$$[1, \infty) \ni t \mapsto |z_\sigma(tw)|$$

is monotonously decreasing.

Proof. We have

$$\mathbb{C}_{[-2,2]}^\sigma = \{z \in \mathbb{C} : |\operatorname{Re}(z)| > 2 \text{ or } \sigma \operatorname{Im}(z) \geq 0\},$$

which immediately implies that, for $w \in \mathbb{C}_{[-2,2]}^\sigma$, one has

$$[1, \infty) w \subseteq \mathbb{C}_{[-2,2]}^\sigma. \quad (15)$$

We will show that the function

$$[1, \infty) \ni t \mapsto |z_\sigma(tw)|^2$$

is monotonously decreasing, which is equivalent to the statement claimed above. For this, we have to consider three cases:

- (1) $w \notin (-2, 2)$
- (2) $w \in (-2, 2) \setminus \{0\}$
- (3) $w = 0$

We will first assume that $w \notin (-2, 2)$. Then $tw \neq \pm 2$ for every $t \in (1, \infty)$, which implies that the function $(1, \infty) \ni t \mapsto z_\sigma(tw)$ is differentiable. Then, for every $t \in (1, \infty)$, by Lemma 2.5(a), one has

$$w = \frac{d}{dt}(tw) = \frac{d}{dt} [z_\sigma(tw) + z_\sigma(tw)^{-1}] = [1 - z_\sigma(tw)^{-2}] \frac{d}{dt} z_\sigma(tw),$$

i.e.

$$\begin{aligned} \frac{d}{dt} z_\sigma(tw) &= \frac{w}{1 - z_\sigma(tw)^{-2}} = z_\sigma(tw) \frac{w}{z_\sigma(tw) - z_\sigma(tw)^{-1}} \\ &= z_\sigma(tw) \frac{w[z_\sigma(tw) - z_\sigma(tw)^{-1}]}{|z_\sigma(tw) - z_\sigma(tw)^{-1}|^2}. \end{aligned}$$

Now, using again Lemma 2.5(a), one has

$$\begin{aligned} &w[z_\sigma(tw) - z_\sigma(tw)^{-1}] \\ &= [z_\sigma(tw) + z_\sigma(tw)^{-1}] [z_\sigma(tw) - z_\sigma(tw)^{-1}] \\ &= |z_\sigma(tw)|^2 - |z_\sigma(tw)|^{-2} + z_\sigma(tw)^{-1} \overline{z_\sigma(tw)} - \overline{z_\sigma(tw)^{-1}} z_\sigma(tw) \\ &= |z_\sigma(tw)|^2 - |z_\sigma(tw)|^{-2} + 2i \operatorname{Im}(z_\sigma(tw)^{-1} \overline{z_\sigma(tw)}), \end{aligned}$$

so

$$\operatorname{Re} \left(w \overline{[z_\sigma(tw) - z_\sigma(tw)^{-1}]} \right) = |z_\sigma(tw)|^2 - |z_\sigma(tw)|^{-2}.$$

This yields

$$\begin{aligned} \frac{d}{dt} |z_\sigma(tw)|^2 &= \frac{d}{dt} \left(z_\sigma(tw) \cdot \overline{z_\sigma(tw)} \right) \\ &= \left(\frac{d}{dt} z_\sigma(tw) \right) \overline{z_\sigma(tw)} + \overline{\left(\frac{d}{dt} z_\sigma(tw) \right)} z_\sigma(tw) \\ &= 2 \operatorname{Re} \left(\left(\frac{d}{dt} z_\sigma(tw) \right) \overline{z_\sigma(tw)} \right) \\ &= 2 \operatorname{Re} \left(z_\sigma(tw) \frac{w \overline{[z_\sigma(tw) - z_\sigma(tw)^{-1}]} \overline{z_\sigma(tw)}}{|z_\sigma(tw) - z_\sigma(tw)^{-1}|^2} \right) \\ &= 2 \left| \frac{z_\sigma(tw)}{z_\sigma(tw) - z_\sigma(tw)^{-1}} \right|^2 \operatorname{Re} \left(w \overline{[z_\sigma(tw) - z_\sigma(tw)^{-1}]} \right) \\ &= 2 \left| \frac{z_\sigma(tw)}{z_\sigma(tw) - z_\sigma(tw)^{-1}} \right|^2 (|z_\sigma(tw)|^2 - |z_\sigma(tw)|^{-2}). \end{aligned} \quad (16)$$

Finally, by (15) and Lemma 2.5(b),(c), we have $|z_\sigma(tw)| \leq 1$, which implies that

$$|z_\sigma(tw)|^2 - |z_\sigma(tw)|^{-2} \leq 0$$

and therefore, by Equation (16), we also have $\frac{d}{dt} |z_\sigma(tw)|^2 \leq 0$.

We now consider the case $w \in (-2, 2) \setminus \{0\}$. In this case, we have $\frac{2w}{|w|} = \pm 2$, so, by our discussion for the case $w \notin (-2, 2)$, the function

$$[1, \infty) \ni t \mapsto \left| z_\sigma \left(t \frac{2w}{|w|} \right) \right|^2$$

is monotonously decreasing. Replacing $\frac{2t}{|w|}$ by t , this shows that the function

$$\left[\frac{2}{|w|}, \infty \right) \ni t \mapsto |z_\sigma(tw)|^2$$

is monotonously decreasing. It therefore remains to show that the function

$$\left[1, \frac{2}{|w|} \right] \ni t \mapsto |z_\sigma(tw)|^2$$

is also monotonously decreasing. In fact, on this interval it is constant, since

$$\left[1, \frac{2}{|w|} \right] w \subseteq [-2, 2]$$

and therefore, by Lemma 2.5(b), for $t \in \left[1, \frac{2}{|w|} \right]$, one has

$$|z_\sigma(tw)|^2 = 1^2 = 1.$$

Finally, we have to consider the case $w = 0$. This case is trivial, since, for every $t \in [1, \infty)$, one has

$$|z_\sigma(tw)|^2 = |z_\sigma(0)|^2,$$

so the function

$$[1, \infty) \ni t \mapsto |z_\sigma(tw)|^2$$

is constant and therefore monotonously decreasing. $\square$

Proposition A.2. *Let $\sigma \in \{\pm\}$ and $a, b \in \mathbb{R}$ with $0 < a < b$. Then*

$$\left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| \leq \frac{b}{a} + \sqrt{\left(\frac{b}{a}\right)^2 - 1} \quad \forall w \in \mathbb{C}_{[-2b, 2b]}^\sigma$$

with equality holding, if and only if $w \in \{-2b, 2b\}$.

Proof. For $x > 2$, one has

$$\sqrt{x-2}^\sigma \sqrt{x+2}^\sigma = \sqrt{x-2} \sqrt{x+2} = \sqrt{x^2 - 4}$$

and for $x < -2$, one has

$$\sqrt{x-2}^\sigma \sqrt{x+2}^\sigma = \left(\sigma i \sqrt{-(x-2)}\right) \left(\sigma i \sqrt{-(x+2)}\right) = -\sqrt{x^2 - 4}.$$

For $x \in \mathbb{R} \setminus [-2, 2]$, we therefore get

$$\sqrt{x-2}^\sigma \sqrt{x+2}^\sigma = \operatorname{sgn}(x) \sqrt{x^2 - 4},$$

so

$$z_\sigma(x) = \frac{x - \sqrt{x-2}^\sigma \sqrt{x+2}^\sigma}{2} = \frac{x - \operatorname{sgn}(x) \sqrt{x^2 - 4}}{2} = \operatorname{sgn}(x) \frac{|x| - \sqrt{x^2 - 4}}{2}. \quad (17)$$

This yields

$$\left| z_\sigma\left(\frac{\pm 2b}{a}\right) \right| = \frac{\left| \frac{\pm 2b}{a} \right| - \sqrt{\left(\frac{\pm 2b}{a}\right)^2 - 4}}{2} = \frac{b}{a} - \sqrt{\left(\frac{b}{a}\right)^2 - 1},$$

so

$$\frac{1}{\left| z_\sigma\left(\frac{\pm 2b}{a}\right) \right|} = \frac{1}{\frac{b}{a} - \sqrt{\left(\frac{b}{a}\right)^2 - 1}} = \frac{b}{a} + \sqrt{\left(\frac{b}{a}\right)^2 - 1}.$$

We therefore have to show that, for $w \in \mathbb{C}_{[-2b, 2b]}^\sigma$, one has

$$\left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| \leq \frac{1}{\left| z_\sigma\left(\frac{\pm 2b}{a}\right) \right|} \quad (18)$$

with equality holding, if and only if $w = \pm 2b$. We do this in five steps by showing that:

(1) (18) holds stictly for $w \in [-2a, 2a]$.

(2) (18) holds for $w \in [-2b, -2a] \cup [2a, 2b]$, with equality holding, if and only if $w = \pm 2b$.

- (3) (18) holds strictly for $w \in \mathbb{R} \setminus [-2b, 2b]$.
(4) (18) holds strictly for $w \in \mathbb{C}_\sigma$.
(5) (18) holds strictly for $w \in \mathbb{C}_{[-2b, 2b]}^\sigma \cap \mathbb{C}_{-\sigma}$.

We now prove these statements:

- (1) For $w \in [-2a, 2a]$, we have $\frac{w}{a}, \frac{w}{b} \in [-2, 2]$ and one has $|\operatorname{Re}(\frac{\pm 2b}{a})| = \frac{2b}{a} > 2$, so, by Lemma 2.5(b),(c), we get

$$\left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| = \frac{1}{1} < \frac{1}{\left| z_\sigma\left(\frac{\pm 2b}{a}\right) \right|}.$$

- (2) By Equation (17), we have

$$\begin{aligned} \frac{d}{dx} |z_\sigma(x)| &= \frac{d}{dx} \frac{|x| - \sqrt{x^2 - 4}}{2} = \frac{\operatorname{sgn}(x) - \frac{x}{\sqrt{x^2 - 4}}}{2} \\ &= \frac{\operatorname{sgn}(x)}{\sqrt{x^2 - 4}} \frac{\sqrt{x^2 - 4} - |x|}{2} = -\frac{\operatorname{sgn}(x)}{\sqrt{x^2 - 4}} |z_\sigma(x)|. \end{aligned} \quad (19)$$

The function

$$\mathbb{R} \setminus (-2, 2) \ni x \mapsto |z_\sigma(x)|$$

is therefore strictly monotonously decreasing on $[2, \infty)$ and strictly monotonously increasing on $(-\infty, -2]$. For $w \in [-2b, -2a] \cup [2a, 2b]$, this implies that

$$\left| z_\sigma\left(\frac{w}{a}\right) \right| \geq \left| z_\sigma\left(\frac{\pm 2b}{a}\right) \right|, \quad (20)$$

with equality holding, if and only if $w = \pm 2b$.

For $w \in [-2b, -2a] \cup [2a, 2b]$, we have $\frac{w}{b} \in [-2, 2]$, which, by Lemma 2.5(b), implies

$$\left| z_\sigma\left(\frac{w}{b}\right) \right| = 1.$$

Together with (20), this yields

$$\left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| = \frac{1}{\left| z_\sigma\left(\frac{w}{a}\right) \right|} \leq \frac{1}{\left| z_\sigma\left(\frac{\pm 2b}{a}\right) \right|},$$

with equality holding, if and only if $w = \pm 2b$.

- (3) By Equation (19), for every $w \in \mathbb{R} \setminus [-2b, 2b]$, we get

$$\begin{aligned} \frac{d}{dw} \left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| &= \frac{-\frac{\operatorname{sgn}(\frac{w}{b})}{b\sqrt{(\frac{w}{b})^2 - 4}} |z_\sigma\left(\frac{w}{b}\right)| |z_\sigma\left(\frac{w}{a}\right)| + \frac{\operatorname{sgn}(\frac{w}{a})}{a\sqrt{(\frac{w}{a})^2 - 4}} |z_\sigma\left(\frac{w}{a}\right)| |z_\sigma\left(\frac{w}{b}\right)|}{\left| z_\sigma\left(\frac{w}{a}\right) \right|^2} \\ &= \operatorname{sgn}(w) \left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| \left[\frac{1}{\sqrt{w^2 - 4a^2}} - \frac{1}{\sqrt{w^2 - 4b^2}} \right]. \end{aligned}$$

The function

$$\mathbb{R} \setminus (-2b, 2b) \ni w \mapsto \left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right|$$

is therefore strictly monotonously decreasing on $[2b, \infty)$ and strictly monotonously increasing on $(-\infty, -2b]$. For $w \in \mathbb{R} \setminus [-2b, 2b]$, by Lemma 2.5(h), we therefore have

$$\left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| < \left| \frac{z_\sigma\left(\frac{\pm 2b}{b}\right)}{z_\sigma\left(\frac{\pm 2b}{a}\right)} \right| = \frac{|z_\sigma(\pm 2)|}{|z_\sigma\left(\frac{\pm 2b}{a}\right)|} = \frac{1}{|z_\sigma\left(\frac{\pm 2b}{a}\right)|}.$$

(4) By Equation (3), we have

$$\begin{aligned} \lim_{|w| \rightarrow \infty} \left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| &= \lim_{|w| \rightarrow \infty} \left| \frac{\frac{w}{a} + \sqrt{\frac{w}{a} - 2}^\sigma \sqrt{\frac{w}{a} - 2}^\sigma}{\frac{w}{b} + \sqrt{\frac{w}{b} - 2}^\sigma \sqrt{\frac{w}{b} - 2}^\sigma} \right| \\ &= \frac{b}{a} \lim_{|w| \rightarrow \infty} \left| \frac{w + \sqrt{w - 2a}^\sigma \sqrt{w - 2a}^\sigma}{w + \sqrt{w - 2b}^\sigma \sqrt{w - 2b}^\sigma} \right| = \frac{b}{a} \\ &< \frac{b}{a} + \sqrt{\left(\frac{b}{a}\right)^2 - 1} = \frac{1}{|z_\sigma\left(\frac{\pm 2b}{a}\right)|}. \end{aligned}$$

By the maximum modulus principle, for $w \in \mathbb{C}_\sigma$, we therefore have

$$\left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| < \max_{w \in \mathbb{R}} \left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| = \frac{1}{|z_\sigma\left(\frac{\pm 2b}{a}\right)|},$$

using (1), (2) and (3) in the last step.

(5) Using statement (4) for $-\sigma$ instead of σ , for $w \in \mathbb{C}_{[-2b, 2b]}^\sigma \cap \mathbb{C}_{-\sigma}$, by Lemma 2.5(i), we get

$$\left| \frac{z_\sigma\left(\frac{w}{b}\right)}{z_\sigma\left(\frac{w}{a}\right)} \right| = \left| \frac{z_{-\sigma}\left(\frac{w}{b}\right)}{z_{-\sigma}\left(\frac{w}{a}\right)} \right| < \frac{1}{|z_{-\sigma}\left(\frac{\pm 2b}{a}\right)|} = \frac{1}{|z_\sigma\left(\frac{\pm 2b}{a}\right)|}. \quad \square$$

We now can prove Proposition 2.7(g):

Proof. One has $\tau_A = 1$, if and only if

$$a_1 = a_2 = \dots = a_L,$$

which implies that

$$Z_\sigma(E) = \text{diag} \left(z_\sigma\left(\frac{E}{a_1}\right), \dots, z_\sigma\left(\frac{E}{a_L}\right) \right) = z_\sigma\left(\frac{E}{a_1}\right) \mathbf{1}_L.$$

This immediately yields

$$\|Z_\sigma(E)\| \cdot \|Z_\sigma(E)^{-1}\| = \left| z_\sigma\left(\frac{E}{a_1}\right) \right| \cdot \left| z_\sigma\left(\frac{E}{a_1}\right)^{-1} \right| = 1 = \tau_A$$

for every $E \in \mathbb{C}_{\{-2a_1, 2a_1\}}^\sigma = \mathbb{C}_{[-2a_1, 2a_1] \setminus (-2a_L, 2a_L)}^\sigma$. We therefore, from now on, can assume that $\tau_A > 1$, i.e. that $a_1 > a_L$.

We first prove the statement for $E \in \mathbb{C}_{[-2a_1, 2a_1]}^\sigma$. Then, by Lemma A.1, we have

$$\left| z_\sigma \left(\frac{E}{a_L} \right) \right| \leq \dots \leq \left| z_\sigma \left(\frac{E}{a_1} \right) \right|.$$

Since

$$Z_\sigma(E) = \text{diag} \left(z_\sigma \left(\frac{E}{a_1} \right), \dots, z_\sigma \left(\frac{E}{a_L} \right) \right),$$

this yields

$$\|Z_\sigma(E)\| = \left| z_\sigma \left(\frac{E}{a_1} \right) \right| \quad \text{and} \quad \|Z_\sigma(E)^{-1}\| = \left| z_\sigma \left(\frac{E}{a_L} \right) \right|^{-1}.$$

By Proposition A.2, we then get

$$\|Z_\sigma(E)\| \cdot \|Z_\sigma(E)^{-1}\| = \left| \frac{z_\sigma \left(\frac{E}{a_1} \right)}{z_\sigma \left(\frac{E}{a_L} \right)} \right| \leq \frac{a_1}{a_L} + \sqrt{\left(\frac{a_1}{a_L} \right)^2 - 1},$$

with equality holding, if and only if $E \in \{-2a_1, 2a_1\}$. The statement then follows since

$$\|A\| = a_1 \quad \text{and} \quad \|A^{-1}\| = \frac{1}{a_L}$$

and therefore

$$\eta_A = \|A\| \cdot \|A^{-1}\| = \frac{a_1}{a_L},$$

which implies

$$\tau_A = \eta_A + \sqrt{\eta_A^2 - 1} = \frac{a_1}{a_L} + \sqrt{\left(\frac{a_1}{a_L} \right)^2 - 1}.$$

It remains to show that the statement also holds on the set

$$\begin{aligned} \mathbb{C}_{[-2a_1, 2a_1] \setminus (-2a_L, 2a_L)}^\sigma \setminus \mathbb{C}_{[-2a_1, 2a_1]}^\sigma &= \{E \in \mathbb{C} : \sigma \operatorname{Im}(E) < 0 \text{ and } |\operatorname{Re}(E)| < 2a_L\} \\ &\subseteq \mathbb{C}_{[-2a_1, 2a_1]}^{-\sigma} \setminus \{-2a_1, 2a_1\}. \end{aligned}$$

On this set, by what we have already seen, one has

$$\|Z_{-\sigma}(E)\| \cdot \|Z_{-\sigma}(E)^{-1}\| < \tau_A.$$

The statement then follows since $|\operatorname{Re}(E)| < 2a_L$, by Proposition 2.7(f), implies

$$Z_\sigma(E) = Z_{-\sigma}(E)^{-1}$$

and therefore

$$\|Z_\sigma(E)\| \cdot \|Z_\sigma(E)^{-1}\| = \|Z_{-\sigma}(E)^{-1}\| \cdot \|Z_{-\sigma}(E)\| < \tau_A. \quad \square$$

B Admissible kernels

The goal of this section is to prove Proposition 3.6. We start with the following lemmata:

Lemma B.1. *Let $\sigma \in \{\pm\}$ and $E \in \mathbb{C}_a^\sigma$. Then, for $n, j \in \mathbb{Z}$, one has*

$$\left\| \phi_+^{E,\sigma}(n)^{-1} K_\phi^\sigma(E, n, j) \right\| \cdot \left\| \phi_+^{E,\sigma}(j) \right\| \leq \frac{3 \|A^{-1}\|}{|E| \|A\|^{-1} + 1} |j - n| \alpha^\sigma(E)^{\max\{|j-n|, |j|\}}$$

and, for $n \in \mathbb{N}_0$, one has

$$\left\| \phi_+^{E,\sigma}(n)^{-1} K_\phi^\sigma(E, n, j) \right\| \cdot \left\| \phi_+^{E,\sigma}(j) \right\| \leq 3 \|A^{-1}\| |j| \alpha^\sigma(E)^{|j|}.$$

Proof. If $j \leq n$, we have $K_\phi^\sigma(E, n, j) = 0$ and the statement is trivial. We can therefore assume that $j > n$. By Lemma 2.11(c), we get

$$\begin{aligned} \left\| \phi_+^{E,\sigma}(n)^{-1} s^{E,\sigma}(j - n) \right\| &= \left\| Z_\sigma^{-n} \sum_{k=1}^{j-n} Z_\sigma^{2k-(j-n)-1} \right\| = \left\| Z_\sigma Z_\sigma^{-j} \sum_{k=1}^{j-n} Z_\sigma^{2(k-1)} \right\| \\ &\leq \|Z_\sigma\| \cdot \|Z_\sigma^{-j}\| \cdot \sum_{k=1}^{j-n} \|Z_\sigma\|^{2(k-1)} \\ &\leq \|Z_\sigma\| \cdot \|Z_\sigma^{-j}\| \cdot (j - n) \max\{1, \|Z_\sigma\|^{2(j-n-1)}\} \\ &\leq \|Z_\sigma\| \cdot |j - n| (\max\{1, \|Z_\sigma\|\})^{2|j-n|} \cdot \|Z_\sigma^{-j}\|. \end{aligned}$$

This yields

$$\begin{aligned} &\left\| \phi_+^{E,\sigma}(n)^{-1} K_\phi^\sigma(E, n, j) \right\| \cdot \left\| \phi_+^{E,\sigma}(j) \right\| \\ &= \left\| \phi_+^{E,\sigma}(n)^{-1} A^{-1} s^{E,\sigma}(j - n) \right\| \cdot \left\| \phi_+^{E,\sigma}(j) \right\| \\ &\leq \|A^{-1}\| \cdot \left\| \phi_+^{E,\sigma}(n)^{-1} s^{E,\sigma}(j - n) \right\| \cdot \left\| \phi_+^{E,\sigma}(j) \right\| \\ &\leq \|A^{-1}\| \cdot \|Z_\sigma\| \cdot |j - n| (\max\{1, \|Z_\sigma\|\})^{2|j-n|} \cdot \|Z_\sigma^{-j}\| \cdot \|Z_\sigma^j\| \\ &\leq \|A^{-1}\| \cdot \frac{3}{|E| \|A\|^{-1} + 1} \cdot |j - n| (\max\{1, \|Z_\sigma\|\})^{2|j-n|} \cdot (\|Z_\sigma\| \cdot \|Z_\sigma^{-1}\|)^{|j|} \\ &\leq \frac{3 \|A^{-1}\|}{|E| \|A\|^{-1} + 1} \cdot |j - n| \alpha^\sigma(E)^{\max\{|j-n|, |j|\}}. \end{aligned}$$

The second statement then follows since

$$\frac{3 \|A^{-1}\|}{|E| \|A\|^{-1} + 1} \leq 3 \|A^{-1}\|$$

and, for $j > n \geq 0$, one has

$$|j - n| = j - n \leq j = |j|.$$

□

Lemma B.2. *Let $\sigma \in \{\pm\}$ and $E \in \mathbb{C}_{[-2a_1, 2a_1]}^\sigma$. Then, for $n, j \in \mathbb{N}$, one has*

$$\left\| \psi_+^{E, \sigma}(n)^{-1} \tilde{K}_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \leq \|A^{-1}C^E\| |j| \alpha^\sigma(E)^{|j|}$$

and, if $j \leq n$, one has

$$\begin{aligned} & \left\| \psi_+^{E, \sigma}(n)^{-1} \tilde{K}_\psi^\sigma(E, n, j)(\mathbf{1}_L - P_1^E) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ & \leq \|A^{-1}C^E\| \left[\|Z_\sigma(\mathbf{1}_L - P_-^E - P_1^E)\|^{2(n-j)} + \frac{1}{n} \right] |j| \alpha^\sigma(E)^{|j|}. \end{aligned}$$

Proof. For $j > n$, we have

$$\begin{aligned} \left\| \psi_+^{E, \sigma}(n)^{-1} \tilde{K}_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| &= \left\| A^{-1}C^E \phi_+^{E, \sigma}(j) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ &\leq \|A^{-1}C^E\| \|Z_\sigma^j\| |j| \|Z_\sigma^{-j}\| \\ &\leq \|A^{-1}C^E\| |j| (\|Z_\sigma\| \cdot \|Z_\sigma^{-1}\|)^{|j|} \\ &\leq \|A^{-1}C^E\| |j| \alpha^\sigma(E)^{|j|}. \end{aligned}$$

For $j \leq n$, we get

$$\begin{aligned} & \left\| \psi_+^{E, \sigma}(n)^{-1} \tilde{K}_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ &= \left\| A^{-1}C^E \psi_+^{E, \sigma}(n)^{-1} \phi_+^{E, \sigma}(n) \psi_+^{E, \sigma}(j) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ &\leq \|A^{-1}C^E\| \left\| Z_\sigma^{2n-j} \left[(\mathbf{1}_L - P_-^E) + \frac{j}{n} P_-^E \right] \right\| \|Z_\sigma^{-j}\| \\ &\leq \|A^{-1}C^E\| \|Z_\sigma\|^{2(n-j)} (\|Z_\sigma\| \cdot \|Z_\sigma^{-1}\|)^j \\ &\leq \|A^{-1}C^E\| |j| \alpha^\sigma(E)^{|j|}, \end{aligned}$$

using in the last step that, by ...[TODO], for $E \in \mathbb{C}_{[-2a_1, 2a_1]}^\sigma$, one has $\|Z_\sigma\| \leq 1$. Similarly, we have

$$\begin{aligned} & \left\| \psi_+^{E, \sigma}(n)^{-1} \tilde{K}_\psi^\sigma(E, n, j)(\mathbf{1}_L - P_1^E) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ &= \left\| A^{-1}C^E \psi_+^{E, \sigma}(n)^{-1} \phi_+^{E, \sigma}(n) \psi_+^{E, \sigma}(j)(\mathbf{1}_L - P_1^E) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ &\leq \|A^{-1}C^E\| \left\| Z_\sigma^{2n-j} \left[(\mathbf{1}_L - P_-^E) + \frac{j}{n} P_-^E \right] (\mathbf{1}_L - P_1^E) \right\| \|Z_\sigma^{-j}\| \\ &\leq \|A^{-1}C^E\| \left\| Z_\sigma^{2(n-j)} (\mathbf{1}_L - P_-^E - P_1^E) + \frac{j}{n} P_-^E \right\| (\|Z_\sigma\| \cdot \|Z_\sigma^{-1}\|)^j \\ &\leq \|A^{-1}C^E\| \left[\|Z_\sigma(\mathbf{1}_L - P_-^E - P_1^E)\|^{2(n-j)} + \frac{j}{n} \|P_-^E\| \right] \alpha^\sigma(E)^j \\ &\leq \|A^{-1}C^E\| \left[\|Z_\sigma(\mathbf{1}_L - P_-^E - P_1^E)\|^{2(n-j)} + \frac{1}{n} \right] |j| \alpha^\sigma(E)^{|j|}. \quad \square \end{aligned}$$

Lemma B.3. *Let $\sigma \in \{\pm\}$ and $E \in \mathbb{C}_{[-2a_1, 2a_1]}^\sigma$. Then, for $n, j \in \mathbb{N}$, one has*

$$\left\| \psi_+^{E, \sigma}(n)^{-1} K_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \leq \|A^{-1}\| \left[\|C^E\| + 1 \right] |j| \max \left\{ j, \alpha^\sigma(E)^{|j|} \right\}.$$

and, if $j \leq n$, one has

$$\begin{aligned} & \left\| \psi_+^{E, \sigma}(n)^{-1} K_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ & \leq \|A^{-1} C^E\| \left[\left\| Z_\sigma(\mathbf{1}_L - P_-^E - P_1^E) \right\|^{2(n-j)} + \frac{1}{n} \right] |j| \alpha^\sigma(E)^{|j|}. \end{aligned}$$

Proof. We first show that, for $n \in \mathbb{N}_0$, one has

$$\left\| \psi_+^{E, \sigma}(n)^{-1} K_\phi^{-\sigma}(E, n, j) P_1^E \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \leq \|A^{-1}\| |j| \max \left\{ j, \alpha^\sigma(E)^{|j|} \right\}.$$

If $j \leq n$, we have $K_\phi^{-\sigma}(E, n, j) = 0$ and the statement is trivial. We can therefore assume that $j > n$. By ...[TODO], we have

$$\psi_+^{E, \sigma}(n)^{-1} P_1^E = \phi_+^{E, \sigma}(n) = \phi_-^{E, -\sigma}(n) = \phi_+^{E, -\sigma}(n)^{-1}.$$

By Lemma 2.11(c), we then get

$$\begin{aligned} \left\| \psi_+^{E, \sigma}(n)^{-1} s^{E, -\sigma}(j - n) P_1^E \right\| &= \left\| \phi_+^{E, -\sigma}(n)^{-1} s^{E, -\sigma}(j - n) P_1^E \right\| \\ &= \left\| Z_{-\sigma}^{-n} \sum_{k=1}^{j-n} Z_{-\sigma}^{2k-(j-n)-1} \right\| \\ &= \left\| \sum_{k=1}^{j-n} Z_{-\sigma}^{2k-j-1} P_1^E \right\| \leq \sum_{k=1}^{j-n} \|Z_{-\sigma} P_1^E\|^{2k-j-1} = j - n. \end{aligned}$$

This yields

$$\begin{aligned} & \left\| \psi_+^{E, \sigma}(n)^{-1} K_\phi^{-\sigma}(E, n, j) P_1^E \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ &= \left\| \psi_+^{E, \sigma}(n)^{-1} A^{-1} s^{E, -\sigma}(j - n) P_1^E \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ &\leq \|A^{-1}\| \cdot |j - n| \cdot \max \left\{ j, \|Z_\sigma^{-1}\|^j \right\} \\ &\leq \|A^{-1}\| \cdot |j - n| \cdot \max \left\{ j, (\|Z_\sigma\| \cdot \|Z_\sigma^{-1}\|)^j \right\} \\ &\leq \|A^{-1}\| \cdot |j| \cdot \max \left\{ j, \alpha^\sigma(E)^{|j|} \right\}, \end{aligned}$$

assuming in the penultimate step that $P_1^E \neq 0$ (otherwise, the statement is trivial) and therefore $\|Z_\sigma\| \geq 1$. Together with Lemma B.2, this yields

$$\begin{aligned} & \left\| \psi_+^{E, \sigma}(n)^{-1} K_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ &\leq \left[\left\| \psi_+^{E, \sigma}(n)^{-1} K_\psi^{-\sigma}(E, n, j) P_1^E \right\| + \left\| \psi_+^{E, \sigma}(n)^{-1} \tilde{K}_\psi^\sigma(E, n, j)(\mathbf{1} - P_1^E) \right\| \right] \cdot \left\| \psi_+^{E, \sigma}(j) \right\| \\ &\leq \|A^{-1} C^E\| |j| \alpha^\sigma(E)^{|j|} + \|A^{-1}\| \cdot |j| \cdot \max \left\{ j, \alpha^\sigma(E)^{|j|} \right\} \\ &\leq \|A^{-1}\| \cdot [\|C^E\| + 1] \cdot |j| \cdot \max \left\{ j, \alpha^\sigma(E)^{|j|} \right\}. \end{aligned}$$

The second statement follows immediately from Lemma B.2 and the fact that, for $j \leq n$, one has $K_\phi^{-\sigma}(E, n, j) = 0$. $\square$

We finish this section with a proof of Proposition 3.6:

Proof. (a) We have

$$\begin{aligned} (H_0 - E)K_\phi^\sigma(E, \cdot, j)(n) &= \delta_j(n)AK_\phi^\sigma(E, n-1, n) \\ &= \delta_j(n)AA^{-1}s^{E,\sigma}(n - (n-1)) = \delta_j(n) \cdot \mathbf{1}_L \end{aligned} \quad (21)$$

and

$$\phi_+^{E,\sigma} \in \mathcal{G}_L(H_0, E).$$

Further, by Lemma B.1, for $N \in \mathbb{N}_0$, one has

$$\begin{aligned} &\sum_{j=N}^{\infty} \sup_{\substack{E \in O_\alpha^\sigma \\ n \geq N}} \left\| \phi_+^{E,\sigma}(n)^{-1} K_\phi^\sigma(E, n, j) V(j) \phi_+^{E,\sigma}(j) \right\| \\ &\leq \sum_{j=N}^{\infty} \sup_{\substack{E \in O_\alpha^\sigma \\ n \geq N}} \left\| \phi_+^{E,\sigma}(n)^{-1} K_\phi^\sigma(E, n, j) \right\| \|V(j)\| \left\| \phi_+^{E,\sigma}(j) \right\| \\ &\leq \sum_{j=N}^{\infty} \sup_{\substack{E \in O_\alpha^\sigma \\ n \geq N}} \|V(j)\| 3 \|A^{-1}\| |j| \alpha^\sigma(E)^{|j|} \\ &\leq 3 \|A^{-1}\| \sum_{j=N}^{\infty} \|V(j)\| |j| \alpha^{|j|} < \infty \end{aligned}$$

and

$$\begin{aligned} &\sum_{j=N}^{\infty} \sup_{E \in O_\alpha^\sigma} \left\| \phi_+^{E,\sigma}(n)^{-1} K_\phi^\sigma(E, n, j) V(j) \phi_+^{E,\sigma}(j) \right\| \\ &= \sum_{j=n+1}^{\infty} \sup_{E \in O_\alpha^\sigma} \left\| \phi_+^{E,\sigma}(n)^{-1} K_\phi^\sigma(E, n, j) V(j) \phi_+^{E,\sigma}(j) \right\| \\ &\leq \sum_{j=n+1}^{\infty} \sup_{E \in O_\alpha^\sigma} \left\| \phi_+^{E,\sigma}(n)^{-1} K_\phi^\sigma(E, n, j) \right\| \|V(j)\| \left\| \phi_+^{E,\sigma}(j) \right\| \\ &\leq \sum_{j=n+1}^{\infty} \sup_{E \in O_\alpha^\sigma} \|V(j)\| 3 \|A^{-1}\| |j| \alpha^\sigma(E)^{|j|} \\ &\leq 3 \|A^{-1}\| \sum_{j=n+1}^{\infty} \|V(j)\| |j| \alpha^{|j|} \xrightarrow{n \rightarrow \infty} 0. \end{aligned}$$

(b) We have

$$\begin{aligned}
& A\phi_+^{E,\sigma}(n+1)\psi_+^{E,\sigma}(n) + A\phi_+^{E,\sigma}(n)\psi_+^{E,\sigma}(n-1) - E\phi_+^{E,\sigma}(n)\psi_+^{E,\sigma}(n) \\
&= AZ_\sigma [(1 - P_\equiv^E) + nP_\equiv^E] + AZ_\sigma [(1 - P_\equiv^E) + (n-1)P_\equiv^E] - E [(1 - P_\equiv^E) + nP_\equiv^E] \\
&= (2AZ_\sigma - E)(1 - P_\equiv^E) + ((2n-1)AZ_\sigma - En)P_\equiv^E \\
&= A [Z_\sigma - Z_\sigma^{-1}] (1 - P_\equiv^E) - AP_\equiv^E = A(C^E)^{-1}.
\end{aligned}$$

This yields

$$\begin{aligned}
& (H_0 - E)\tilde{K}_\psi^\sigma(E, \cdot, j)(n) \\
&= \delta_j(n) [AK_\psi^\sigma(E, n+1, n) + AK_\psi^\sigma(E, n-1, n) - EK_\psi^\sigma(E, n, n)] \\
&= \delta_j(n) \cdot \mathbf{1}_L.
\end{aligned}$$

Together with Equation (21), this yields

$$\begin{aligned}
(H_0 - E)K_\psi^\sigma(E, \cdot, j) &= (H_0 - E) [K_\psi^{-\sigma}(E, \cdot, j)P_1^E + \tilde{K}_\psi^\sigma(E, \cdot, j)(\mathbf{1}_L - P_1^E)] \\
&= [\delta_j(n) \cdot \mathbf{1}_L P_1^E + \delta_j(n) \cdot \mathbf{1}_L (\mathbf{1}_L - P_1^E)] = \delta_j(n) \cdot \mathbf{1}_L.
\end{aligned}$$

Also, we have

$$\psi_+^{E,\sigma} \in \mathcal{G}_L(H_0, E).$$

If $\alpha > 1$, there exists $N \in \mathbb{N}_0$ such that, for every $j \geq N$, one has

$$j \leq \alpha^j.$$

Then, for $j \geq N$, by Lemma B.3, we have

$$\begin{aligned}
\left\| \psi_+^{E,\sigma}(n)^{-1} K_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E,\sigma}(j) \right\| &\leq \|A^{-1}\| [\|C^E\| + 1] |j| \max \left\{ j, \alpha^\sigma(E)^{|j|} \right\} \\
&\leq \|A^{-1}\| [\|C^E\| + 1] |j| \alpha^{|j|}.
\end{aligned}$$

If $P_1^E = 0$, by Lemma B.2, we have

$$\begin{aligned}
\left\| \psi_+^{E,\sigma}(n)^{-1} K_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E,\sigma}(j) \right\| &= \left\| \psi_+^{E,\sigma}(n)^{-1} \tilde{K}_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E,\sigma}(j) \right\| \\
&\leq \|A^{-1}C^E\| |j| \alpha^\sigma(E)^{|j|} \\
&\leq \|A^{-1}\| \|C^E\| |j| \alpha^{|j|}.
\end{aligned}$$

If $P_1^E = \mathbf{1}_L$, by Lemma B.1, we have

$$\begin{aligned}
\left\| \psi_+^{E,\sigma}(n)^{-1} K_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E,\sigma}(j) \right\| &= \left\| \phi_+^{E,-\sigma}(n)^{-1} K_\phi^{-\sigma}(E, n, j) \right\| \cdot \left\| \phi_+^{E,-\sigma}(j) \right\| \\
&\leq 3 \|A^{-1}\| |j| \alpha^\sigma(E)^{|j|} \\
&\leq \|A^{-1}\| 3 |j| \alpha^{|j|}.
\end{aligned}$$

In any case, we get

$$\begin{aligned}
\left\| \psi_+^{E,\sigma}(n)^{-1} K_\psi^\sigma(E, n, j) \right\| \cdot \left\| \psi_+^{E,\sigma}(j) \right\| &\leq \|A^{-1}\| [\|C^E\| + 1] |j| \max \left\{ j, \alpha^\sigma(E)^{|j|} \right\} \\
&\leq \|A^{-1}\| [\|C^E\| + 3] |j| \alpha^{|j|}.
\end{aligned}$$

Then

$$\begin{aligned}
& \sum_{j=N}^{\infty} \sup_{n \geq N} \left\| \psi_+^{E,\sigma}(n)^{-1} K_{\psi}^{\sigma}(E, n, j) V(j) \psi_+^{E,\sigma}(j) \right\| \\
& \leq \sum_{j=N}^{\infty} \sup_{n \geq N} \left\| \psi_+^{E,\sigma}(n)^{-1} K_{\psi}^{\sigma}(E, n, j) \right\| \left\| \psi_+^{E,\sigma}(j) \right\| \|V(j)\| \\
& \leq \sum_{j=N}^{\infty} \sup_{n \geq N} \|A^{-1}\| [\|C^E\| + 3] |j| \alpha^{|j|} \|V(j)\| \\
& \leq \|A^{-1}\| [\|C^E\| + 3] \sum_{j=N}^{\infty} \|V(j)\| |j| \alpha^{|j|} < \infty. \tag{22}
\end{aligned}$$

Finally, we have to show that

$$\sum_{j=N}^{\infty} \left\| \psi_+^{E,\sigma}(n)^{-1} K_{\psi}^{\sigma}(E, n, j) V(j) \psi_+^{E,\sigma}(j) \right\| \xrightarrow{n \rightarrow \infty} 0.$$

So let $\varepsilon > 0$. We choose $M \geq N$ such that

$$\|A^{-1}\| [\|C^E\| + 3] \sum_{j=M+1}^{\infty} \|V(j)\| |j| \alpha^{|j|} < \frac{\varepsilon}{2}.$$

Then, as in (22), one shows that

$$\begin{aligned}
& \sum_{j=M+1}^{\infty} \left\| \psi_+^{E,\sigma}(n)^{-1} K_{\psi}^{\sigma}(E, n, j) V(j) \psi_+^{E,\sigma}(j) \right\| \\
& \leq \|A^{-1}\| [\|C^E\| + 3] \sum_{j=M+1}^{\infty} \|V(j)\| |j| \alpha^{|j|} < \frac{\varepsilon}{2} \quad \forall n \in \mathbb{N}_0.
\end{aligned}$$

Thus, it remains to show that

$$\sum_{j=N}^M \left\| \psi_+^{E,\sigma}(n)^{-1} K_{\psi}^{\sigma}(E, n, j) V(j) \psi_+^{E,\sigma}(j) \right\| \xrightarrow{n \rightarrow \infty} 0.$$

For $n > M$, by Lemma B.3, one has

$$\begin{aligned}
& \sum_{j=N}^M \left\| \psi_+^{E,\sigma}(n)^{-1} K_\psi^\sigma(E, n, j) V(j) \psi_+^{E,\sigma}(j) \right\| \\
&= \sum_{j=N}^M \left\| \psi_+^{E,\sigma}(n)^{-1} \left[K_\psi^{-\sigma}(E, n, j) P_1^E + \tilde{K}_\psi^\sigma(E, n, j) (\mathbf{1}_L - P_1^E) \right] V(j) \psi_+^{E,\sigma}(j) \right\| \\
&= \sum_{j=N}^M \left\| \psi_+^{E,\sigma}(n)^{-1} \left[0 \cdot P_1^E + \tilde{K}_\psi^\sigma(E, n, j) (\mathbf{1}_L - P_1^E) \right] V(j) \psi_+^{E,\sigma}(j) \right\| \\
&\leq \sum_{j=N}^M \left\| \psi_+^{E,\sigma}(n)^{-1} \tilde{K}_\psi^\sigma(E, n, j) (\mathbf{1}_L - P_1^E) \right\| \|V(j)\| \left\| \psi_+^{E,\sigma}(j) \right\| \\
&\leq \sum_{j=N}^M \|V(j)\| \|A^{-1}C^E\| \left[\|Z_\sigma(\mathbf{1}_L - P_-^E - P_1^E)\|^{2(n-j)} + \frac{1}{n} \right] |j| \alpha^\sigma(E)^{|j|} \\
&\leq \|A^{-1}C^E\| \left[\|Z_\sigma(\mathbf{1}_L - P_-^E - P_1^E)\|^{2(n-M)} + \frac{1}{n} \right] \sum_{j=N}^M \|V(j)\| |j| \alpha^{|j|} \xrightarrow{n \rightarrow \infty} 0,
\end{aligned}$$

using that

$$\|Z_\sigma(\mathbf{1}_L - P_-^E - P_1^E)\| < 1.$$

□

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